

A) Environment Ecology

Environment, Ecology, Ecosystem, Energy Flow, Food Chain and Food Web, Biogeochemical Cycles, Terrestrial and Aquatic Ecosystems, Biotic Interactions etc.

1] With reference to the planet Earth, consider the following statements: [2025]

- I. Rain forests produce more oxygen than that produced by oceans.
- II. Marine phytoplankton and photosynthetic bacteria produce about 50% of world's oxygen.
- III. Well-oxygenated surface water contains several folds higher oxygen than that in atmospheric air.

Which of the statements given above is/are correct?

- a) I and II
- b) II only
- c) I and III
- d) None of the above statements is correct]

2] Consider the following: [2024]

- 1. Carabid beetles
- 2. Centipedes
- 3. Flies
- 4. Termites
- 5. Wasps

Parasitoid species are found in how many of the above kind of organisms?

- a) Only two
- b) Only three
- c) Only four
- d) All five

Thus, it appears that b) Only three is correct answer.

3] Consider the following plants: [2024]

- 1. Groundnut
- 2. Horse-gram
- 3. Soybean

How many of the above belong to the pea family?

- a) Only one
- b) Only two
- c) All three
- d) None

4] The organisms "Cicada, Froghopper and Pond skater" are: [2024]

- a) Birds
- b) Fish
- c) Insects
- d) Reptiles

5] Consider the following pairs: [2024]

Country- Animal round in its natural habitat

- 1. Brazil- Indri
- 2. Indonesia- Elk
- 3. Madagascar- Bonobo

How many of the pairs given above are correctly matched?

- a) Only one
- b) Only two
- c) All three
- d) None

6] Consider the following statements: [2024]

- 1. Lions do not have a particular breeding season,
- 2. Unlike most other big cats, cheetahs do not roar.
- 3. Unlike male lions, male leopards do not proclaim their territory by scent marking.

Which of the statements given above are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

7] Which one of the following shows a unique relationship with an insect that has co-evolved with it and that is the only insect that can pollinate this tree? [2024]

- a) Fig
- b) Mahua
- c) Sandalwood
- d) Silk cotton

8] Consider the following: [2024]

- 1. Butterflies
- 2. Fish
- 3. Frogs

How many of the above have poisonous species among them?

- a) Only one
- b) Only two
- c) All three
- d) None

9] Which of the following organisms perform waggle dance for others of their kin to indicate the direction and the distance to a source of their food? [2023]

- a) Butterflies
- b) Dragonflies
- c) Honeybees
- d) Wasps

10] Which of the following makes a tool with a stick to scrape insects from a hole in a tree or a log of wood? [2023]

- a) Fishing cat b) Orangutan
- c) Otter d) Sloth bear

11] Consider following statements: [2023]

1. Some mushrooms have medicinal properties.
 2. Some mushrooms have psycho-active properties.
 3. Some mushrooms have insecticidal properties.
 4. Some mushrooms have bioluminescent properties.
- How many of the above statements are correct?

- a) Only one b) Only two
- c) Only three d) All four

12] "If rainforests and tropical' forests are the lungs of the Earth, then surely. 'wetlands function as its kidneys." Which one of the following functions of wetlands best reflects the above statement? [2022]

- a) The water cycle in wetlands involves surface runoff, subsoil percolation and evaporation.
- b) Algae form the nutrient base upon which fish, crustaceans, molluscs, birds, reptiles and mammals thrive.
- c) Wetlands play a vital role in maintaining sedimentation balance and soil stabilization.
- d) Aquatic plants absorb heavy metals and excess nutrients.

13] Consider the following kinds of organisms: [2021]

- 1) Copepods 2) Cyanobacteria
- 3) Diatoms 4) Foraminifera

Which of the above are primary producers in the food chains of oceans?

- a) 1 and 2 b) 2 and 3
- c) 3 and 4 d) 1 and 4

14] In case of which one of the following biogeochemical cycles the weathering of rocks is the main source of release of nutrient to enter the cycle? [2021]

- a) Carbon cycle b) Nitrogen cycle
- c) Phosphorus cycle d) Sulphur cycle

15] Which of the following are detritivores? [2021]

- 1) Earthworms 2) Jellyfish
- 3) Millipedes 4) Seahorses
- 5) Woodlice

Select the correct answer using the code given below.

- a) 1, 2 & 4 only b) 2, 3, 4 & 5 only
- c) 1, 3 & 5 only d) 1, 2, 3, 4 & 5

16] Which of the following have species that can establish symbiotic relationship with other organisms? [2021]

- 1) Cnidarians 2) Fungi 3) Protozoa

Select the correct answer using the code given below.

- a) 1 and 2 only b) 2 and 3 only
- c) 1 and 3 only d) 1, 2 and 3

17] Consider the following statements: [2019]

1. Some species of turtles are herbivores.
2. Some species of fish are herbivores.
3. Some species of marine mammals are herbivores.
4. Some species of snakes are viviparous.

Which of the statements given above is/are correct?

- a) 1 and 3 only b) 2, 3 and 4 only
- c) 2 and 4 only d) 1, 2, 3 and 4

18] Which of the following leaf modifications occur(s) in the desert areas to inhibit water loss? [2018]

1. Hard and waxy leaves
2. Tiny leaves
3. Thorns instead of leaves

Select the correct answer using the code given below:

- a) 2 and 3 only b) 2 only
- c) 3 only d) 1, 2 and 3

19] Due to some reasons, if there is a huge fall in the population of species of butterflies, what could be its likely consequence/consequences? [2017]

1. Pollination of some plants could be adversely affected.
2. There could be a drastic increase in the fungal infections of some cultivated plants.
3. It could lead to a fall in the population of some species of wasps, spiders and birds.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

20] Which one of the following is the best description of the term 'ecosystem'? [2015]

- a) A community of organisms interacting with one another
- b) That part of the Earth which is inhabited by living organisms
- c) A community of organisms together with the environment in which they live.
- d) The flora and fauna of a geographical area.

21] Lichens, which are capable of initiating ecological succession even on a bare rock, are actually a symbiotic association of: [2014]

- a) Algae and bacteria
- b) Algae and fungi
- c) Bacteria and fungi
- d) Fungi and mosses

22] Which one of the following is the correct sequence of a food chain? [2014]

- a) Diatoms-Crustaceans-Herrings
- b) Crustaceans-Diatoms-Herrings
- c) Diatoms-Herrings-Crustaceans
- d) Crustaceans-Herrings-Diatoms

23] Which of the following adds/ads carbon dioxide to the carbon cycle on the planet Earth? [2014]

- 1. Volcanic action
- 2. Respiration
- 3. Photosynthesis
- 4. Decay of organic matter

Select the correct answer using the code given below.

- a) 1 and 3 only
- b) 2 only
- c) 1, 2 and 4 only
- d) 1, 2, 3 and 4

24] Which of the following leaf modifications occurs/occur in desert areas to inhibit water loss? [2013]

- 1. Hard and waxy leaves
- 2. Tiny leaves or no leaves
- 3. Thorns instead of leaves

Select the correct answer using the codes given below.

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 1, 2 and 3

25] With reference to food chains in ecosystems, consider the following statements: [2013]

- 1. A food chain illustrates the order in which a chain of organisms feed upon each other.
- 2. Food chains are found within the population of a species.
- 3. A food chain illustrates the numbers of each organism which are eaten by others.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 1, 2 and 3
- d) None

26] Which of the following adds/ adds nitrogen to the soil? [2013]

- 1. Excretion of Urea by animals
- 2. Burning of coal by man
- 3. Death of vegetation

Select the correct answer using the codes given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

27] Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms? [2013]

- a) Ecotone
- b) Ecological niche
- c) Habitat
- d) Home range

28] In the grasslands, trees do not replace the grasses as a part of an ecological succession because of___? [2013]

- a) Insects and fungi
- b) Limited sunlight and paucity of nutrients
- c) Water limits and fire
- d) None of the above

29] Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity? [2013]

- a) Keystone species b) Allopatric species
c) Sympatric species d) Threatened species

39] Estuaries possess distinct blooms of excessive growth of a pigmented dinoflagellates. These blooms are called: [1998]

- a) Red tides
b) Sea tides
c) Black tides
d) Sea flowers

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
b	b	c	c	d	a	a	c	c	b	d	d	b	c	c	d	d	d	c	c

21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39
b	a	c	d	a	c	b	c	c	b	d	c	c	b	a	d	b	a	a

B) Biodiversity in India

Fauna, Flora and Indian Wildlife Diversity

1] Regarding Peacock tarantula (Gooty tarantula), consider the following statements: [2025]

- I. It is an omnivorous crustacean.
II. Its natural habitat in India is only limited to some forest areas.
III. In its natural habitat, it is an arboreal species.

Which of the statements given above is/are correct?

- a) I only
b) I and III
c) II only
d) II and III

2] Consider the following fruits: [2025]

- I. Papaya
II. Pineapple
III. Guava

How many of the above were introduced in India by the Portuguese in the sixteenth and seventeenth centuries?

- a) Only one
b) Only two
c) All the three
d) None

3] Consider the following pairs: [2025]

Plant – Description

- I. Cassava : Woody shrub
II. Ginger : Herb with pseudostem
III. Malabar spinach : Herbaceous climber
IV. Mint : Annual shrub
V. Papaya : Woody shrub

How many of the above pairs are correctly matched?

- a) Only two
b) Only three

- c) Only four
d) All the five

4] Consider the following statements: [2024]

Statement I: The Indian Flying Fox is placed under the "vermin" category in the Wild Life (Protection) Act, 1972.

Statement-II: The Indian Flying Fox feeds on the blood of other animals.

Which one of the following is correct in respect of the above statements?

- a) Statement-I and Statement-II are correct and Statement-II explains Statement-I
b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
c) Statement-I is correct, but Statement-II is incorrect
d) Statement-I is incorrect, but Statement-II is correct

5] Consider the following: [2024]

1. Cashew
2. Papaya
3. Red sanders

How many of the above trees are actually native to India?

- a) Only one
b) Only two
c) All three
d) None

6] Consider the following trees: [2023]

1. Jackfruit (Artocarpus heterophyllus)
2. Mahua (Madhuca indica)

How many of the above are deciduous trees?

- 7] Consider the following statements: [2023]

Which of the following is correct in respect of the above statements?

- 8] Consider the following Fauna: [2023]

- How many of the above are generally nocturnal or most active after sunset?

- 9] Consider the following statements regarding the Indian squirrels: [2023]

- How many of the above statements are correct?

- 10] Certain species of which one of the following organisms are well known as cultivators of fungi? [2022]

- 11] With reference to “Gucchi” sometimes mentioned in the news, consider the following statements: [2022]

1. It is a fungus.
2. It grows in some Himalayan Forest areas.
3. It is commercially cultivated in the Himalayan foothills of north-eastern India.

Which of the statements given above is/are correct?

- a) 1 only b) 3 only
c) 1 and 2 d) 2 and 3

12] Which of the following is not a bird? [2022]

- a) Golden Mahseer b) Indian Nightjar
c) Spoonbill d) White Ibisw

13] "Biorock technology" is talked about in which one of the following situations? [2022]

- a) Restoration of damaged coral reefs
- b) Development of building materials using plant residues
- c) Identification of areas for exploration/extraction of shale gas
- d) Providing salt licks for wild animals in forests/protected areas

14] Consider the following animals: [2021]

- 1) Hedgehog
- 2) Marmot
- 3) Pangolin

To reduce the chance of being captured by predators which of the above organisms rolls up/roll up and protects/protect its/their vulnerable parts?

- a) 1 and 2 b) 2 only
c) 3 only d) 1 and 3

15] Which one of the following is used in preparing a natural mosquito repellent? [2021]

- a) Congress grass b) Elephant grass
c) Lemon grass d) Nut grass

16] Which one of the following is a filter feeder?
[2021]

- a) Catfish b) Octopus
c) Oyster d) Pelican

17] With reference to Indian elephants, consider the following statements: [2020]

1. The leader of an elephant group is a female.
2. The maximum gestation period can be 22 months.
3. An elephant can normally go on calving till the age of 40 years only.
4. Among the States in India, the highest elephant population is in Kerala.

Which of the statements given above is/are correct?

- a) 1 and 2 only b) 2 and 4 only
c) 3 only d) 1, 3 and 4 only

18] With reference to India's biodiversity, Ceylon frogmouth, Coppersmith barbet, Gray-chinned miniyet and White-throated redstart are: [2020]

- a) Birds b) Primates
c) Reptiles d) Amphibians

19] Which of the following are the most likely places to find the musk deer in its natural habitat? [2020]

1. Askot Wildlife Sanctuary
2. Gangotri National Park
3. Kishanpur Wildlife Sanctuary
4. Manas National Park

Select the correct answer using the code given below:

- a) 1 and 2 only b) 2 and 3 only
c) 3 and 4 only d) 1 and 4 only

20] Which one of the following groups of plants was domesticated in the 'New World' and introduced into the 'Old World'? [2019]

- a) Tobacco, cocoa and rubber
b) Tobacco, cotton and rubber
c) Cotton, coffee and sugarcane
d) Rubber, coffee and wheat

21] Consider the following statements: [2019]

1. Asiatic lion is naturally found in India only.
2. Double-humped camel is naturally found in India only.
3. One-horned rhinoceros is naturally found in India only.

Which of the statements given above is/are correct?

- a) 1 only b) 2 only
c) 1 and 3 only d) 1, 2 and 3

22] Consider the following pairs: [2019]

Wildlife: Naturally found in

1. Blue-finned Mahseer: Cauvery River
2. Irrawaddy Dolphin: Chambal River
3. Rusty-spotted Cat: Eastern Ghats

Which of the pairs given correctly matched?

- a) 1 and 2 only b) 2 and 3 only
c) 1 and 3 only d) 1, 2 and 3

23] Recently, there was a growing awareness in our country about the importance of Himalayan nettle (*Girardinia diversifolia*) because it is found to be a sustainable source of: [2019]

- a) Anti-malarial drug b) Biodiesel
c) Pulp for paper industry d) Textile fibre

24] Consider the following statements: [2018]

1. Most of the world's coral reefs are in tropical waters.
2. More than one-third of the world's coral reefs are located in the territories of Australia, Indonesia and Philippines.
3. Coral reefs host far more number of animal phyla than those hosted by tropical rainforests.

Which of the statements given above is/are correct?

- a) 1 and 2 only b) 3 only
c) 1 and 3 only d) 1, 2 and 3

25] Why is a plant called *Prosopis juliflora* often mentioned in news? [2018]

- a) Its extract is widely used in cosmetics.
b) It tends to reduce the widely biodiversity in the area in which it grows.
c) Its extract is used in the synthesis of pesticides.
d) None of the above

26] If you want to see gharials in their natural habitat, which one of the following is the best place to visit? [2017]

- a) Bhitarkanika Mangroves
b) Chambal River
c) Pulicat Lake

d) Deepor Beel

27] What is/are unique about 'Kharai Camel', a breed found in India? [2016]

1. It is capable of swimming up to three kilometres in seawater.
2. It survives by grazing on mangroves.
3. It lives in the wild and cannot be domesticated.

Select the correct answer using the code given below.

- a) 1 and 2 only b) 3 only
c) 1 and 3 only d) 1, 2 and 3

28] Recently, our scientists have discovered a new and distinct species of banana plant which attains a height of about 11 metres and has orange-coloured fruit pulp. In which part of India has it been discovered? [2016]

- a) Andaman Islands
b) Anaimalai Forests
c) Maikala Hills
d) Tropical rain forests of northeast

29] With reference to 'Red Sanders', sometimes seen in the news, consider the following statements: [2016]

1. It is a tree species found in a part of South India.
2. It is one of the most important trees in the tropical rain forest areas of South India.

Which of the statements given above is/are correct?

- a) 1 only b) 2 only
c) Both 1 and 2 d) Neither 1 nor 2

30] In which of the following regions of India are you most likely to come across the 'Great Indian Hornbill' in its natural habitat? [2016]

- a) Sand deserts of northwest India
b) Higher Himalayas of Jammu and Kashmir
c) Salt marshes of western Gujarat
d) Western Ghats

31] With reference to 'dugong', a mammal found in India, which of the following statements is/are correct? [2015]

1. It is a herbivorous marine animal.
2. It is found along the entire coast of India

3. It is given legal protection under Schedule 1 of the Wildlife (Protection) Act, 1972.

Select the correct answer using the code given below.

- a) 1 and 2 b) 2 only
c) 1 and 3 d) 3 only

32] Which one of the following is the national aquatic animal of India? [2015]

- a) Saltwater crocodile b) Olive ridley turtle
c) Gangetic dolphin d) Gharial

33] Which one of the following regions of India has a combination of mangrove forest, evergreen forest and deciduous forest? [2015]

- a) North Coastal Andhra Pradesh
b) South-West Bengal
c) Southern Saurashtra
d) Andaman and Nicobar Islands

34] If you travel through the Himalayas, you are likely to see which of the following plants naturally growing there? [2014]

1. Oak
2. Rhododendron
3. Sandalwood

Select the correct answer using the code given below.

- a) 1 and 2 only b) 3 only
c) 1 and 3 only d) 1, 2 and 3

35] The most important strategy for the conservation of biodiversity together with traditional human life is the establishment of: [2014]

- a) Biosphere b) Reserves
c) National parks d) Wildlife sanctuaries

36] In India, cluster bean (Guar) is traditionally used as a vegetable or animal feed, but recently the cultivation of this has assumed significance. Which one of the following statements is correct in this context? [2014]

- a) The oil extracted from seeds is used in the manufacture of biodegradable plastics.
b) The gum made from its seeds is used in the extraction of shale gas

- c) The leaf extract of this plant has the properties of anti-histamines.
d) It is a source of high quality biodiesel.

37] With reference to Neem tree, consider the following statements: [2014]

1. Neem oil can be used as a pesticide to control the proliferation of some species of insects and mites.
2. Neem seeds are used in the manufacture of biofuels and hospital detergents.
3. Neem oil has applications in pharmaceutical industry.

Which of the statements given above is/are correct?

- a) 1 and 2 only b) 3 only
c) 1 and 3 only d) 1, 2 and 3

38] Other than poaching, what are the possible reasons for the decline in the population of Gangas River Dolphins? [2014]

1. Construction of dams and barrages on rivers
2. Increase in the population of crocodiles in rivers.
3. Getting trapped in fishing nets accidentally
4. Use of synthetic fertilizers and other agricultural chemicals in crop-fields in the vicinity of rivers

Select the correct answer using the code given below.

- a) 1 and 2 only b) 2 and 3 only
c) 1, 3 and 4 only d) 1, 2, 3 and 4

39] If you walk through countryside, you are likely to see some birds stalking alongside the cattle to seize the insects disturbed by their movement through grasses. Which one of the following is/are such bird/birds? [2014]

1. Painted Stork
2. Common Myna
3. Black-necked Crane

Select the correct answer using the code given below.

- a) 1 and 2 b) 2 only
c) 2 and 3 d) 3 only

40] Which of the following have coral reefs? [2014]

1. Andaman and Nicobar Islands
2. Gulf of Kachchh
3. Gulf of Mannar
4. Sunderbans

Select the correct answer using the code given below.

- a) 1, 2 and 3 only b) 2 and 4 only
c) 1 and 3 only d) 1, 2, 3 and 4

41] Consider the following fauna of India____: [2013]

1. Gharial
2. Leatherback turtle
3. Swamp deer

Which among the above is / are endangered?

- a) 1 and 2 only b) 3 Only
c) 1, 2 & 3 d) None

42] In which of the following states is lion-tailed macaque found in its natural Habitat? [2013]

1. Tamil Nadu
2. Kerala
3. Karnataka
4. Andhra Pradesh

Select the correct answer using the codes given below.

- a) 1, 2 and 3 only b) 2 only
c) 1, 3 and 4 only d) 1, 2, 3, and 4

43] Consider the following: [2013]

1. Star tortoise
2. Monitor lizard
3. Pygmy hog
4. Spider monkey

Which of the above are naturally found in India?

- a) 1, 2 and 3 only b) 2 and 3 only
c) 1 and 4 only d) 1, 2, 3, and 4

44] The Government of India encourages the cultivation of 'sea buckthorn '. What is the importance of this plant? [2012]

1. It helps in controlling soil erosion and in preventing desertification.
2. It is a rich source of biodiesel.
3. It has nutritional value and is well-adapted to live in cold areas of high altitudes.
4. Its timber is of great commercial value.

Which of the statements given above is/are correct?

- a) 1 Only b) 2, 3 & 4 Only
c) 1 & 3 Only d) 1, 2, 3 & 4

45] Which one of the following groups of animals belongs to the category of endangered species? [2012]

- a) Great Indian Bustard, Musk Deer, Red Panda and Asiatic Wild Ass
- b) Kashmir Stag, Cheetai, Blue Bull and Great Indian Bustard
- c) Snow Leopard, Swamp Deer, Rhesus Monkey and Saras (Crane)
- d) Lion-tailed Macaque, Blue Bull, Hanuman Langur and Cheetal

46] What is the difference between the antelopes Oryx and Chiru? [2012]

- a) Oryx is adapted to live in hot and arid areas whereas Chiru is adapted to live in steppes and semi-desert areas of cold high mountains
- b) Oryx is poached for its antlers whereas Chiru is poached for its musk
- c) Oryx exists in western India only whereas Chiru exists in north-east India only
- d) None of the statements (a), (b) and 'c' given above is correct

47] Consider the following: [2012]

- 1. Black-necked crane
- 2. Cheetah
- 3. Flying squirrel
- 4. Snow leopard

Which of the above are naturally found in India?

- a) 1, 2 and 3 only b) 1, 3 and 4 only
- c) 2 and 4 only d) 1, 2, 3 and 4

48] What would happen if phytoplankton of an ocean is completely destroyed for some reason? [2012]

- 1. The ocean as a carbon sink would be adversely affected
- 2. The food chains in the ocean would be adversely affected.
- 3. The density of ocean water would drastically decrease

Select the correct answer:

- a) 1 & 2 Only b) 2 Only
- c) 3 Only d) 1, 2 & 3

49] Vultures which used to be very common in the Indian countryside some years ago are rarely seen nowadays. This is attributed to: [2012]

- a) The destruction of their nesting sites by new invasive species
- b) A drug used by cattle owners for treating their diseased cattle
- c) Scarcity of food available to them
- d) A widespread, persistent and fatal disease among them

50] The acidification of Oceans is increasing. Why is this phenomenon a cause of concern? [2012]

- 1. The growth and survival of calcareous phytoplankton will be adversely affected.
- 2. The growth and survival of coral reefs will be adversely affected.
- 3. Survival of some animals that have phytoplanktonic larvae will be adversely affected.
- 4. The cloud seeding and formation of clouds will be adversely affected

Which among the statements given above is / are correct?

- a) 1, 2 & 3 only b) 2 Only
- c) 1 & 3 Only d) 1, 2, 3 & 4

51] Which of the following can be threats to the biodiversity of a geographical area? [2012]

- 1. Global warming
- 2. Fragmentation of habitat
- 3. Invasion of alien species
- 4. Promotion of vegetarianism

Choose the correct answer:

- a) 1, 2 & 3 b) 2 & 3
- c) 1 & 4 d) 1, 2, 3 and 4

52] Biodiversity forms the basis for human existence in the following ways: [2011]

- 1. Soil formation.
- 2. Prevention of soil erosion.
- 3. Recycling of waste.
- 4. Pollination of crops.

Select the correct answer using the codes given below:

- a) 1, 2, and 3 only b) 2, 3 and 4 only

1. Eastern Himalayas
2. Eastern Mediterranean region

3. North-western Australia

Which of the above is/are Biodiversity Hotspot (s)?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

62] The marine animal called Dugong which is vulnerable to extinction is: [2009]

- a) Amphibian
- b) Bony fish
- c) Shark
- d) Mammal

63] In the contest of Indian wild life, the flying fox is a: [2009]

- a) Bat
- b) Kite
- c) Stork
- d) Vulture

64] Among the following, which one is not an ape? [2008]

- a) Gibbon
- b) Gorilla
- c) Langur
- d) Orangutan

65] Which one of the following is an insectivorous plant? [2008]

- a) Passion flower plant
- b) Pitcher plant
- c) Night queen
- d) Flame of the forest.

66] For which one of the following snakes is the diet mainly composed of other snakes? [2008]

- a) Krait
- b) Russell's viper
- c) Rattlesnake
- d) King cobra

67] In which one of the following kinds of organisms is the phenomenon found wherein the female kills the male after copulation? [2008]

- a) Dragonfly
- b) Honeybee
- c) Spider
- d) Pit viper

68] Consider the following statements: [2008]

1. Salt-water crocodile is found in the Andaman and Nicobar Islands.
2. Shrew and tapir are found in the Western Ghats of the Malabar region.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only

c) Both 1 and 2

d) Neither 1 nor 2

69] Which one of the following is not essentially a species of the Himalayan vegetation? [2008]

- a) Juniper
- b) Mahogany
- c) Silver fur
- d) Spruce

70] Consider the following statements: [2007]

1. In India, Red Panda is naturally found in the Western Himalayas only.
2. In India, Slow Loris lives in the dense forests of the North East.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

71] Dalbergia species is associated with which one of the following? [2007]

- a) Cashew nut
- b) Coffee
- c) Tea
- d) Rosewood

72] The largest coral reef in the world is found near the coast of which one of the following countries? [2007]

- a) Australia
- b) Cuba
- c) Ghana
- d) Philippines

73] Consider the following statements: [2004]

1. Toothless mammals such as pangolins are not found in India.
2. Gibbon is the only ape found in India.

Which of the statements given above is are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

74] Consider the following animals of India: [2003]

1. Crocodile
2. Elephant

Which of these is/are endangered species?

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

75] Consider the following statements: [2003]

1. The common blue green algae Spirogyra and Ulothrix are found in both freshwater ponds and oceans.

2. The Chameleon can look ahead with one eye, and at the same time look behind with another.

Which of these statements is/are correct?

- a) Only 1 b) Only 2
c) Both 1 and 2 d) Neither 1 nor 2

76] Match List I (Mangrove) with List II (State) and select the correct answer using the codes given below the lists: [2002]

List I. (Mangrove) List II(State)

- | | |
|--------------------|-------------------|
| A. Achra Ratnagiri | 1. Karnataka |
| B. Coondapur | 2. Kerala |
| C. Pichavaram | 3. Andhra Pradesh |
| D. Vembanad | 4. Maharashtra |
| | 5. Tamil Nadu |

A B C D

- a) 2 1 5 4
b) 4 5 3 2
c) 2 5 3 4
d) 4 1 5 2

77] The sea coast of which one of the following states has become famous as a nesting place for the giant Olive Ridley turtles from South America? [2002]

- a) Goa b) Gujarat
c) Orissa d) Tamil Nadu

78] A class of animals known as Marsupials is a characteristic feature of: [2001]

- a) Africa b) Australia
c) South America d) South-East Asia

79] 'India has the largest population of the Asian X.

Today, there are just about 20,000 to 25,000 X in their natural habitat spreading across the evergreen forests, dry thorn forests, swamps and grasslands. Their prime habitats are, however, the moist deciduous forests. The X population in India ranges from North-West India where they are found in the forest divisions of Dehradun, Bijnor and Nainital

districts of UP to the Western Ghats in the states of Karnataka and Kerala and in Tamil Nadu. In Central India, their population is distributed in southern Bihar and Orissa. In the East, they are seen in North-Bengal, Assam and a few other states.' The animal 'X' referred to in this quotation is: [1999]

- a) Lion b) Elephant
c) Tiger d) One-horned rhinoceros

80] A tree species in Mauritius failed to reproduce because of the extinction of a fruit-eating bird. Which one of the following was that bird? [1998]

- a) Dove b) Dodo
c) Condor d) Skua

81] Assertion (A): Mangroves are very specialised forest ecosystems of tropical and subtropical regions bordering certain sea coasts.: [1996]

Reason (R): They stabilise the shoreline and act as bulwark against encroachments by sea. In the context of the above two statements, which one of the following is correct?

- a) Both A and R are true and R is the correct explanation of A
b) Both A and R are true but R is not a correct explanation of A
c) A is true but R is false
d) A is false but R is true

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
d	c	b	X	a	b	c	a	b	a	c	a	a	d	c	c	a	a	a	a

21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
a	c	d	d	b	b	a	a	a	d	c	c	d	a	a	b	c	c	b	a

41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
c	a	a	c	a	a	b	a	b	a	a	d	b	b	d	b	c	c	b	d

61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
b	d	a	c	b	d	c	c	b	b	d	a	b	c	b	d	c	b	b	b

81
a

C) Environmental Pollution and Remedies

Air, Water, Soil, Noise Pollution, Radioactive Pollution, e-waste, Acid Rain, Remedies for Pollution, Govt. Initiatives, Bio-remediation etc.

1] Consider the following statements: [2025]

Statement I: Activated carbon is a good and an attractive tool to remove pollutants from effluent streams and to remediate contaminants from various industries.

Statement II: Activated carbon exhibits a large surface area and a strong potential for adsorbing heavy metals.

Statement III: Activated carbon can be easily synthesized from environmental wastes with high carbon content.

Which one of the following is correct in respect of the above statements?

- a) Both Statement II and Statement III are correct and both of them explain Statement I
- b) Both Statement II and Statement III are correct but only one of them explains Statement I
- c) Only one of the Statements II and III is correct and that explains Statement I
- d) Neither Statement II nor Statement III is correct

2] Consider the following: [2025]

I. Cigarette butts

II. Eyeglass lenses

III. Car tyres

How many of them contain plastic?

- a) Only one
- b) Only two
- c) All the three
- d) None

3] With reference to Perfluoroalkyl and polyfluoroalkyl substances (PFAS) that are used in making many consumer products, consider the following statements: [2024]

- 1. PFAS are found to be widespread in drinking water, food and food packaging materials.
- 2. PFAS are not easily degraded in the environment.
- 3. Persistent exposure to PFAS can lead to bioaccumulation in animal bodies.

Which of the statements given above are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

4] Consider the following statements: [2024]

Statement I: Many chewing gums found in the market are considered a source of environmental pollution.

Statement-II: Many chewing gums contain plastic as gum base.

Which one of the following is correct in respect of the above statements?

- a) Statement-I and Statement-II are correct and Statement-II explains Statement-I
- b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
- c) Statement-I is correct, but Statement-II is incorrect
- d) Statement-I is incorrect, but Statement-II is correct

5] According to the Environmental Protection Agency (EPA), which one of the following is the largest source of sulphur dioxide emissions? [2024]

- a) Locomotives using fossil fuels
- b) Ships using fossil fuels
- c) Extraction of metals from ores
- d) Power plants using fossil fuels

6] "Membrane Bioreactors" are often discussed in the context of: [2024]

- a) Assisted reproductive technologies
- b) Drug delivery nanotechnologies
- c) Vaccine production technologies
- d) Wastewater treatment technologies

7] With reference to the role of biofilters in Recirculating Aquaculture System, consider the following statements: [2023]

- 1. Biofilters provide waste treatment by removing uneaten fish feed.
- 2. Biofilters convert ammonia present in fish waste to nitrate.
- 3. Biofilters increase phosphorus as nutrient for fish in water.

How many of the statements given above are correct?

- a) Only one
- b) Only two
- c) All three
- d) None

8] Consider the following statements regarding mercury pollution: [2023]

- 1. Gold mining activity is a source of mercury pollution in the world.
- 2. Coal-based thermal power plants cause mercury pollution.
- 3. There is no known safe level of exposure to mercury.

How many of the above statements are correct?

- a) Only one
- b) Only two
- c) All three
- d) None

9] Consider the following: [2022]

- 1. Carbon monoxide
- 2. Nitrogen oxide
- 3. Ozone
- 4. Sulphur dioxide

Excess of which of the above in the environment is/are the cause(s) of acid rain?

- a) 1, 2 and 3
- b) 2 and 4 only
- c) 4 only
- d) 1, 3 and 4

10] In the context of WHO Air Quality Guidelines, consider the following statements: [2022]

- 1. The 24-hour mean of PM 2.5 should not exceed 15 $\mu\text{g}/\text{m}^3$ and annual mean of PM2.5 should not exceed 5 $\mu\text{g}/\text{m}^3$.
- 2. In a year, the highest levels of ozone pollution occur during the periods of inclement weather.
- 3. PM 10 can penetrate the lung barrier and enter the bloodstream.
- 4. Excessive ozone in the air can trigger asthma.

Which of the statements given above are correct?

- a) 1, 3 and 4
- b) 1 and 4 only
- c) 2, 3 and 4
- d) 1 and 2 only

11] Consider the following statements: [2022]

- 1. Other than those made by humans, nanoparticles do not exist in nature.
- 2. Nanoparticles of some metallic oxides are used in the manufacture of some cosmetics.
- 3. Nanoparticles of some commercial products which enter the environment are unsafe for humans.

Which of the statements given above is/are correct?

- a) 1 only
- b) 3 only

c) 1 and 2

d) 2 and 3

12] Magnetite particles suspected to cause neurodegenerative problems, are generated as environmental pollutants from which of the following? [2021]

- 1) Brakes of motor vehicles
- 2) Engines of motor vehicles
- 3) Microwave stoves within homes
- 4) Power plants
- 5) Telephone lines

Select the correct answer using the code given below.

- a) 1, 2, 3 and 5 only b) 1, 2 and 4 only
c) 3, 4 and 5 only d) 1, 2, 3, 4 and 5

13] 'R2 Code of Practices' constitutes a tool available for promoting the adoption of: [2021]

- a) Environmentally responsible practices in electronics recycling industry
- b) Ecological management of 'Wetlands of International Importance' under the Ramsar
- c) Sustainable practices in the cultivation of agricultural crops in degraded lands
- d) 'Environmental Impact Assessment' in the exploitation of natural resources

14] Why is there a concern about copper smelting plants? [2021]

- 1) They may release lethal quantities of carbon monoxide into environment.
- 2) The copper slag can cause the leaching of some heavy metals into environment.
- 3) They may release sulphur dioxide as a pollutant.

Select the correct answer using the code given below.

- a) 1 and 2 only b) 2 and 3 only
c) 1 and 3 only d) 1, 2 and 3

15] With reference to furnace oil, consider the following statements: [2021]

- 1) It is a product of oil refineries.
- 2) Some industries use it to generate power
- 3) Its use causes sulphur emissions into environment.

Which of the statements given above are correct?

- a) 1 and 2 only b) 2 and 3 only
c) 1 and 3 only d) 1, 2 and 3

16] Which of the following are the reasons/factors for exposure to benzene pollution? [2020]

1. Automobile exhaust
2. Tobacco smoke
3. Wood burning
4. Using varnished wooden furniture
5. Using products made of polyurethane

Select the correct answer using the code given below:

- a) 1, 2 and 3 only b) 2 and 4 only
c) 1, 3 and 4 only d) 1, 2, 3, 4 and 5

17] Consider the following statements: [2020]

1. Coal ash contains arsenic, lead and mercury.
2. Coal-fired power plants release sulphur dioxide and oxides of nitrogen into the environment.
3. High ash content is observed in Indian coal.

Which of the statements given above is/are correct?

- a) 1 only b) 2 and 3 only
c) 3 only d) 1, 2 and 3

18] Steel slag can be the material for which of the following? [2020]

1. Construction of base road
2. Improvement of agricultural soil
3. Production of cement

Select the correct answer using the code given below:

- a) 1 and 2 only b) 2 and 3 only
c) 1 and 3 only d) 1, 2 and 3

19] Why is there a great concern about the 'microbeads' that are released into environment? [2019]

- a) They are considered harmful to marine ecosystems.
- b) They are considered to cause skin cancer in children.
- c) They are small enough to be absorbed by crop plants in irrigated fields.
- d) They are often found to be used as food adulterants.

20] In India, the use of carbofuran, methyl parathion, phorate and triazophos is viewed with apprehension. These chemicals are used as: [2019]

- a) Pesticides in agriculture
- b) Preservatives in processed foods
- c) Fruit-ripening agents
- d) Moisturising agents in cosmetics

21] Consider the following statements: [2019]

The Environment Protection Act, 1986 empowers the Government of India to

1. State the requirement of public participation in the process of environmental protection, and the procedure and manner in which it is sought
2. Lay down the standards for emission or discharge of environmental pollutants from various sources

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

22] As per the Solid Waste Management Rules, 2016 in India, which one of the following statements is correct? [2019]

- a) Waste generator has to segregate waste into five categories.
- b) The Rules are applicable to notified urban local bodies, notified towns and all industrial townships only.
- c) The Rules provide for exact and elaborate criteria for the identification of sites for landfills and waste processing facilities.
- d) It is mandatory on the part of the waste generator that the waste generated in one district cannot be moved to another district.

23] In India, 'extended producer responsibility' was introduced as an important feature in which of the following? [2019]

- a) The Bio-medical Waste (Management and Handling) Rules, 1998
- b) The Recycled Plastic (Manufacturing and Usage) Rules, 1999
- c) The E-Waste (Management and Handling) Rules, 2011
- d) The Food Safety and Standard Regulations, 2011

24] How is the National Green Tribunal (NGT) different from Central Pollution Control Board (CPCB)? [2018]

1. The NGT has been established by an Act and CPCB has been created by an executive order of the government.
2. The NGT provides environmental justice and helps reducing the burden of litigation in the High Courts whereas the CPCB promotes the cleanliness of streams and wells, aims to improve the quality of air in the country.

Which of the statement(s) given below is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

25] In the context of solving pollution problems, what is/are the advantage/advantages of bioremediation technique? [2017]

1. It is a technique for cleaning up pollution by enhancing the same biodegradation process that occurs in nature.
2. Any contaminant with heavy metals such as cadmium and lead can be readily and completely treated by bioremediation using microorganisms.
3. Genetic engineering can be used to create microorganisms specifically designed for bioremediation.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

26] Biological Oxygen Demand (BOD) is a standard criterion for: [2017]

- a) Measuring oxygen levels in blood
- b) Computing oxygen levels in forest ecosystems
- c) Pollution assay in aquatic ecosystems
- d) Assessing oxygen levels in high altitude regions

27] In the cities of our country, which among the following atmospheric gases are normally considered in calculating the value of the Air Quality Index? [2016]

1. Carbon dioxide
2. Carbon monoxide
3. Nitrogen dioxide
4. Sulphur dioxide
5. Methane

Select the correct answer using the code given below.

- | | |
|--------------------|---------------------|
| a) 1, 2 and 3 only | b) 2, 3 and 4 only |
| c) 1, 4 and 5 only | d) 1, 2, 3, 4 and 5 |

28] With reference to bio-toilets used by the Indian Railways, consider the following statements: [2015]

1. The decomposition of human waste in the bio-toilets is initiated by a fungal inoculum.
2. Ammonia and water vapour are the only end products in this decomposition which are released into the atmosphere.

Which of the statements given above is/are correct?

- | | |
|-----------------|--------------------|
| a) 1 only | b) 2 only |
| c) Both 1 and 2 | d) Neither 1 nor 2 |

29] With reference to 'fly ash' produced by the power plants using the coal as fuel, which of the following statements is/are correct? [2015]

1. Fly ash can be used in the production of bricks for building construction
2. Fly ash can be used as a replacement for some of the Portland cement contents of concrete
3. Fly ash is made up of silicon dioxide and calcium oxide only, and does not contain any toxic elements.

Select the correct answer using the code given below

- | | |
|------------|-----------|
| a) 1 and 2 | b) 2 only |
| c) 1 and 3 | d) 3 only |

30] Which of the following are some important pollutants released by the steel industry in India? [2014]

1. Oxides of sulphur
2. Oxide of nitrogen
3. Carbon monoxide
4. Carbon dioxide

Select the correct answer using the code given below.

- | | |
|--------------------|------------------|
| a) 1, 3 and 4 only | b) 2 and 3 only |
| c) 1 and 4 only | d) 1, 2, 3 and 4 |

31] Brominated flame retardants are used in many household products like mattresses and upholstery. Why is there some concern about their use? [2014]

1. They are highly resistant to degradation in the environment.
2. They are able to accumulate in humans and animals.

Select the correct answer using the code given below.

- | | |
|-----------------|--------------------|
| a) 1 only | b) 2 only |
| c) Both 1 and 2 | d) Neither 1 nor 2 |

32] There is some concern regarding the nanoparticles of some chemical elements that are used by the industry in the manufacture of various products. Why? [2014]

1. They can accumulate in the environment and contaminate water and soil.
2. They can enter the food chains.
3. They can trigger the production of free radicals.

Select the correct answer using the code given below.

- | | |
|-----------------|---------------|
| a) 1 and 2 only | b) 3 only |
| c) 1 and 3 only | d) 1, 2 and 3 |

33] Due to improper / indiscriminate disposal of old and used computers or their parts, which of the following are released into the environment as e-waste? [2013]

1. Beryllium
2. Cadmium
3. Chromium
4. Heptachlor
5. Mercury
6. Lead
7. Plutonium

Select the correct answer using the codes given below.

- | |
|----------------------------|
| a) 1, 3, 4, 6, and 7 only |
| b) 1, 2, 3, 5, and 6 only |
| c) 2, 4, 5 and 7 only |
| d) 1, 2, 3, 4, 5, 6, and 7 |

34] Acid rain is caused by the pollution of the environment by? [2013]

- | |
|---------------------------------------|
| a) Carbon dioxide and nitrogen |
| b) Carbon monoxide and carbon dioxide |

- c) Ozone and carbon dioxide
- d) Nitrous oxide and sulphur dioxide

35] Photochemical smog is a resultant of the reaction among? [2013]

- a) NO₂, O₃ and peroxyacetyl nitrate in the presence of sunlight
- b) CO, CO₂ and peroxyacetyl nitrate in the presence of sunlight
- c) CO, CO₂ and NO₂ at low temperature
- d) High concentration of NO₂, O₃ and CO in the evening

36] Which of the following can be found as pollutants in the drinking water in some parts of India? [2013]

- 1. Arsenic
- 2. Sorbitol
- 3. Fluoride
- 4. Formaldehyde
- 5. Uranium

Select the correct answer using the codes given below.

- a) 1 and 3 only
- b) 2, 4 and 5 only
- c) 1, 3, and 5 only
- d) 1, 2, 3, 4, and 5

37] Lead, ingested or inhaled, is a health hazard. After the addition of lead to petrol has been banned, what still are the sources of lead poisoning? [2012]

- 1. Smelting units
- 2. Pens and pencils
- 3. Paints
- 4. Hair oils and cosmetics

Select the correct answer using the codes given below:

- a) 1,2 & 3 only
- b) 1 and 3 only
- c) 2 & 4 only
- d) 1,2,3 & 4

38] There is a concern over the increase in harmful algal blooms in the seawater of India. What could be the causative factors for this phenomenon? [2011]

- 1. Discharge of nutrients from the estuaries.
- 2. Run-off from the land during the monsoon.'
- 3. Upwelling in the seas.

Select the correct answer from the codes given below:

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1,2, and 3

39] Consider the following: [2011]

- 1. Carbon dioxide.
- 2. Oxides of nitrogen.
- 3. Oxides of Sulphur

Which of the above is/are the emission/emissions from coal combustion at thermal power plants?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2, and 3

40] Indiscriminate disposal of used fluorescent electric lamps causes mercury pollution in the environment. Why is mercury used in the manufacture of these lamps? [2010]

- a) A mercury coating on the inside of the lamp makes the light bright white.
- b) When the lamp is switched on, the mercury in the lamp causes the emission of ultra-violet radiations.
- c) When the lamp is switched on, it is the mercury which converts the ultra- violet energy into visible light.
- d) None of the statement given above is correct about the use of mercury in the manufacture of fluorescent lamps.

41] Which bacterial strain, developed from natural isolates by genetic manipulations, can be used for treating oil spills? [2010]

- a) Agrobacterium
- b) Clostridium
- c) Nitrosomonas
- d) Pseudomonas

42] Assertion (A): Coal-based thermal power stations contribute to acid-rain.: [2003]

Reason (R): Oxides of carbon are emitted when the coal burns.

- a) Both A and R are individually true and R is the correct explanation of A
- b) Both A and R are individually true but R is NOT the correct explanation of A
- c) A is true but R is false
- d) A is false but R is true

43] Which one of the following is produced during the formation of photochemical smog? [2003]

- a) Hydrocarbons b) Nitrogen Oxides
c) Ozone d) Methane

44] The water pollution in river is measured by the dissolved amount of: [1998]

- a) Chlorine b) Ozone
c) Nitrogen d) Oxygen

45] Which one of the following fuels causes minimum environmental pollution? [1995]

- a) Diesel b) Coal
c) Hydrogen d) Kerosene

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
a	c	d	a	d	d	b	b	b	b	d	b	a	b	d	a	d	d	a	a

21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
b	c	c	b	c	c	b	d	a	d	c	d	b	d	a	c	b	d	d	b

41	42	43	44	45
d	b	c	d	c

D) Climate Change and Remedies

Climate Change Theoretical Questions

1] The World Bank warned that India could become one of the first places where wet-bulb temperatures routinely exceed 35 °C. Which of the following statements best reflect(s) the implication of the above-said report? [2025]

- I. Peninsular India will most likely suffer from flooding, tropical cyclones and droughts.
II. The survival of animals including humans will be affected as shedding of their body heat through perspiration becomes difficult.

Select the correct answer using the code given below.

- a) I only b) II only
c) Both I and II d) Neither I nor II

2] Consider the following statements: [2025]

Statement I: Circular economy reduces the emissions of greenhouse gases.

Statement II: Circular economy reduces the use of raw materials as inputs.

Statement III: Circular economy reduces wastage in the production process.

Which one of the following is correct in respect of the above statements?

- a) Both Statement II and Statement III are correct and both of them explain Statement I
b) Both Statement II and Statement III are correct but only one of them explains Statement I
c) Only one of the Statements II and III is correct and that explains Statement I
d) Neither Statement II nor Statement III is correct

3. Consider the following statements: [2025]

Statement I: Studies indicate that carbon dioxide emissions from cement industry account for more than 5% of global carbon emissions.

Statement II: Silica-bearing clay is mixed with limestone while manufacturing cement.

Statement III: Limestone is converted into lime during clinker production for cement manufacturing.

Which one of the following is correct in respect of the above statements?

- a) Both Statement II and Statement III are correct and both of them explain Statement I
- b) Both Statement II and Statement III are correct but only one of them explains Statement I
- c) Only one of the Statements II and III is correct and that explains Statement I
- d) Neither Statement II nor Statement III is correct

4] With reference to Direct Air Capture, an emerging technology, which of the following statements is/are correct? [2025]

- I. It can be used as a way of carbon sequestration.
- II. It can be a valuable approach for plastic production and in food processing.
- III. In aviation, it can be a source of carbon for combining with hydrogen to create synthetic low-carbon fuel.

Select the correct answer using the code given below.

- a) I and II only
- b) III only
- c) I, II and III
- d) None of the above statements is correct

5] Consider the following statements: [2025]

- I. Carbon dioxide (CO₂) emissions in India are less than 0.5 t CO₂/capita.
- II. In terms of CO₂ emissions from fuel combustion, India ranks second in Asia-Pacific region.
- III. Electricity and heat producers are the largest sources of CO₂ emissions in India.

Which of the statements given above is/are correct?

- a) I and III only b) II only
- c) II and III only d) I, II and III

6] One of the following regions has the world's largest tropical peatland, which holds about three years' worth of global carbon emissions from fossil fuels; and the possible destruction of which can exert detrimental effect on the global climate. Which one of the following denotes that region? [2024]

- a) Amazon Basin b) Congo Basin
- c) Kikori Basin d) Rio de la Plata Basin

7] Consider the following: [2023]

- 1. Aerosols

- 2. Foam agents
- 3. Fire retardants
- 4. Lubricants

In the making of how many of the above are hydrofluorocarbons used?

- a) Only one b) Only two
- c) Only three d) All four

8] Consider the following activities: [2023]

- 1. Spreading finely ground basalt rock on farmlands extensively
 - 2. Increasing the alkalinity of oceans by adding lime
 - 3. Capturing carbon dioxide released by various industries and pumping it into abandoned subterranean mines in the form of carbonated waters
- How many of the above activities are often considered and discussed for carbon capture and sequestration?

- a) Only one b) Only two
- c) All three d) None

9] Among the following crops, which one is the most important anthropogenic source of both methane and nitrous oxide? [2022]

- a) Cotton b) Rice
- c) Sugarcane d) Wheat

10] What is blue carbon? [2021]

- a) Carbon captured by oceans and coastal ecosystems
- b) Carbon sequestered in forest biomass and agricultural soils
- c) Carbon contained in petroleum and natural gas
- d) Carbon present in atmosphere

11] In rural road construction, the use of which of the following is preferred for ensuring environmental sustainability or to reduce carbon footprint? [2020]

- 1. Copper slag
- 2. Cold mix asphalt technology
- 3. Geotextiles
- 4. Hot mix asphalt technology
- 5. Portland cement

Select the correct answer using the code given below:

- a) 1, 2 and 3 only b) 2, 3 and 4 only

c) 4 and 5 only

d) 1 and 5 only

12] Which of the following statements best describes the term 'Social Cost of Carbon'? [2020]

It is a measure, in monetary value, of the

- a) Long-term damage done by a tonne of CO₂ emissions in a given year.
- b) Requirement of fossil fuels for a country to provide goods and services to its citizens, based on the burning of those fuels.
- c) Efforts put in by a climate refugee to adapt to live in a new place.
- d) Contribution of an individual person to the carbon footprint on the planet Earth.

13] Which of the following statements are correct about the deposits of 'methane hydrate'? [2019]

- 1. Global warming might trigger the release of methane gas from these deposits.
- 2. Large deposits of 'methane hydrate' are found in Arctic Tundra and under the seafloor.
- 3. Methane in atmosphere oxidises to carbon dioxide after a decade or two.

Select the correct answer using the code given below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

14] Consider the following: [2019]

- 1. Carbon monoxide
- 2. Methane
- 3. Ozone
- 4. Sulphur dioxide

Which of the above are released into atmosphere due to the burning of crop/biomass residue?

- a) 1 and 2 only
- b) 2, 3 and 4 only
- c) 1 and 4 only
- d) 1, 2, 3 and 4

15] Consider following statements: [2019]

- 1. Agricultural soils release nitrogen oxides into the environment.
- 2. Cattle release ammonia into the environment.
- 3. Poultry industry releases reactive nitrogen compounds into the environment.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 and 3 only

c) 2 only

d) 1, 2 and 3

16] In the context of proposals to the use of hydrogen-enriched CNG (H-CNG) as fuel for buses in public transport, consider the following statements: [2019]

- 1. The main advantage of the use of H-CNG is the elimination of carbon monoxide emissions.
- 2. H-CNG as fuel reduces carbon dioxide and hydrocarbon emissions.
- 3. Hydrogen up to one-fifth by volume can be blended with CNG as fuel for buses.
- 4. H-CNG makes the fuel less expensive than CNG.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 4 only
- d) 1, 2, 3 and 4

17] Which of the following statements best describes 'carbon fertilization'? [2018]

- a) Increased Plant growth due to increased concentration of carbon dioxide in the atmosphere
- b) Increased temperature of the earth due to increased concentration of carbon dioxide in the atmosphere
- c) Increased acidity of oceans as a result of increased concentration of carbon dioxide in the atmosphere
- d) Adaptation of living beings on Earth to the climate change brought about by the increased concentration of carbon dioxide in the atmosphere

18] The term "sixth mass extinction/sixth extinction" is often mentioned in the news in the context of discussion of: [2018]

- a) Widespread monoculture practices in agriculture and large-scale commercial farming with indiscriminate use of chemicals in many parts of the world that may result in the loss of good native ecosystems.
- b) Fears of possible collision of a meteorite with the Earth in the near future in the manner it happened 65 million years ago that caused the mass extinction of many species including those of dinosaurs.
- c) Large scale cultivation of genetically modified crops in many part of the world and promoting their cultivation in other parts of the world which may

cause the disappearance of good native crop plants and the loss of food biodiversity.

d) Mankind's overexploitation/misuse of natural resources, fragmentation/loss of natural habitats, destruction of ecosystems, pollutions and world climate change.

19] In the context of mitigating the impending global warming due to anthropogenic emissions of carbon dioxide, which of the following can be the potential sites for carbon sequestration? [2017]

1. Abandoned and uneconomic coal seams
2. Depleted oil and gas reservoirs
3. Subterranean deep saline formations

Select the correct answer using the code given below:

- | | |
|-----------------|---------------|
| a) 1 and 2 only | b) 3 only |
| c) 1 and 3 only | d) 1, 2 and 3 |

20] With reference to an organization known as 'Birdlife International' which of the following statements is/are correct? [2015]

1. It is a Global Partnership of Conservation Organizations.
2. The concept of 'biodiversity hotspots' originated from this organization.
3. It identifies the sites known/referred to as 'Important Bird and Biodiversity Areas'.

Select the correct answer using the code given below.

- | | |
|-----------------|-----------------|
| a) 1 only | b) 2 and 3 only |
| c) 1 and 3 only | d) 1, 2 and 3 |

21] With reference to the International Union for Conservation of Nature and Natural Resources (IUCN) and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which of the following statements is/are correct? [2015]

1. IUCN is an organ of the United Nations and CITES is an international agreement between governments
2. IUCN runs thousands of field projects around the world to better manage natural environments.
3. CITES is legally binding on the States that have joined it, but this Convention does not take the place of national laws.

Select the correct answer using the code given below.

- | | |
|-----------------|-----------------|
| a) 1 only | b) 2 and 3 only |
| c) 1 and 3 only | d) 1, 2 and 3 |

22] With reference to a conservation organization called 'Wetlands International', which of the following statements is/are correct? [2014]

1. It is an intergovernmental organization formed by the countries which are signatories to Ramsar Convention.
2. It works at the field level to develop and mobilize knowledge, and use the practical experience to advocate for better policies.

Select the correct answer using the code given below.

- | | |
|-----------------|--------------------|
| a) 1 only | b) 2 only |
| c) Both 1 and 2 | d) Neither 1 nor 2 |

23] If a wetland of international importance is brought under the 'Montreux Record', what does it imply? [2014]

- a) Changes in ecological character have occurred, are occurring or are likely to occur in the wetland as a result of human interference.
- b) The country in which the wetland is located should enact a law to prohibit any human activity within five kilometres from the edge of the wetland.
- c) The survival of the wetland depends on the cultural practices and traditions of certain communities living in its vicinity and therefore the cultural diversity therein should not be destroyed.
- d) It is given the status of 'World Heritage Site'.

24] The scientific view is that the increase in global temperature should not exceed 2 degree Celsius above pre-industrial level. If the global temperature increases beyond 3 degree Celsius above the pre-industrial level, what can be its possible impact/impacts on the world? [2014]

1. Terrestrial biosphere tends toward a net carbon source.
2. Widespread coral mortality will occur.
3. All the global wetlands will permanently disappear.
4. Cultivation of cereals will not be possible anywhere in the world.

Select the correct answer using the code given below:

- a) 1 only b) 1 and 2 only
c) 2, 3 and 4 only d) 1, 2, 3 and 4

25] Consider the following statement: [2012]

Chlorofluorocarbons, known as ozone depleting substances, are used:

1. in the production of plastic foams
2. in the production of tubeless tyres
3. in cleaning certain electronic components
4. as pressurizing agents in aerosol cans

Which of the statements given above is/are correct?

- a) 1,2 and 3 only b) 4 Only
c) 1,3 & 4 Only d) 1,2,3 & 4

26] The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere, because it absorbs: [2012]

- a) The water vapour of the air and retains its heat
b) The ultraviolet part of the solar radiation
c) All the solar radiations
d) Infrared part of solar radiation

27] Consider the following agricultural practices: [2012]

1. Contour bunding
2. Relay cropping
3. Zero tillage

In the context of global climate change, which of the above helps/help in carbon sequestration/storage in the soil?

- a) 1 & 2 Only b) 3 Only
c) 1,2 & 3 d) None of them

28] The formation of ozone hole in the Antarctic region has been a cause of concern. What could be the reason for the formation of this hole? [2011]

- a) Presence of prominent tropo-spheric turbulence; and inflow of chlorofluorocarbons.
b) Presence of prominent polar front and stratospheric clouds; and inflow of chlorofluorocarbons.
c) Absence of polar front and stratospheric clouds; and inflow of methane and chlorofluorocarbons.

d) Increased temperature at polar region due to global warming.

29] Human activities in the recent past have caused the increased concentration of carbon dioxide in the atmosphere, but a lot of it does not remain in the lower atmosphere because of? [2011]

1. Its escape into the outer stratosphere.
2. The photosynthesis by phytoplankton in the oceans.
3. The trapping of air in the polar ice caps.

Which of the statements given above is/are correct?

- a) 1 and 2 b) 2 only
c) 2 and 3 d) 3 only

30] The 'Red Books' published by the international union for conservation of nature and natural resources (IUCN) contain lists of? [2011]

1. Endemic plant and animal species present in the biodiversity hotspots.
2. Threatened plant and animal species.
3. Protected sites for conservation of nature and natural resources in various countries.

Select the correct answer using the codes given below:

- a) 1 and 3 b) 2 only
c) 2 and 3 d) 3 only

31] Consider the following which can be found in the ambient atmosphere: [2010]

1. Soot
2. Sulphur hexafluoride
3. Water vapour

Which of the above contribute to the warming up of the atmosphere?

- a) 1 and 2 only b) 3 only
c) 2 and 3 only d) 1, 2 and 3

32] Due to their extensive rice cultivation, some regions may be contributing to global warming. To what possible reason / reasons is this attributable? [2010]

1. The anaerobic conditions associated with rice cultivation cause the emission of methane.

2. When nitrogen-based fertilizers are used, nitrous oxide is emitted from the cultivated soil.

Which of the statements given above is/are correct?

- a) 1 only b) 2 only
c) Both 1 and 2 d) Neither 1 nor 2

33] Consider the following: [2008]

1) Rice fields

2) Coal mining

3) Domestic animals

4) Wetlands

Which of the above are sources of methane, a major greenhouse gas?

- a) 1 and 4 only b) 2 and 3 only
c) 1,2 and 3 only d) 1,2,3 and 4

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
b	a	b	c	c	b	c	c	b	a	a	a	d	d	d	b	a	d	d	c

21	22	23	24	25	26	27	28	29	30	31	32	33
b	b	a	b	c	d	b	b	b	b	d	c	d

E) Indian Initiatives for Environment Protection

Various Govt. and Non-Govt. Initiatives, for Biodiversity and Environmental Protection, for Climate Change, Ozone Depletion etc. e.g. Wildlife Protection Act, Protected Areas, Various Projects etc. Biosphere Reserves, National Parks, Ramsar Convention and Sites, Environment Protection Act etc.

1] Consider the following statements: [2023]

Once the Central Government notifies an area as a 'Community Reserve'

1. The Chief Wildlife Warden of the State becomes the governing authority of such forest
2. Hunting is not allowed in such area
3. People of such area are allowed to collect non-timber forest produce
4. People of such area are allowed traditional agricultural practices

How many of the above statements are correct?

- a) Only one b) Only two
c) Only three d) All Four

2] With reference to 'Scheduled Areas' in India, consider the following statements : [2023]

1. Within a State, the notification of an area as Scheduled Area takes place through an order of the President.

2. The largest administrative unit forming the Scheduled Area is the District and the lowest is the cluster of villages in the Block.

3. The Chief Ministers of the concerned State are required to submit annual reports to the Union Home Ministry on the administration of Schedule Areas in the States.

How many of the above statements are correct?

- a) Only one b) Only two
c) All three d) None

3] Consider the following statements: [2023]

1. In India, the Biodiversity Management Committees are key to the realization of the objectives of the Nagoya Protocol.

2. The Biodiversity Management Committees have important functions in determining access and benefit sharing, including the power to levy collection fees on the access of biological resources within its jurisdiction.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 or 2

4] Which one of the following has been constituted under the Environment (Protection) Act, 1986? [2022]

- a) Central Water Commission
- b) Central Ground Water Board
- c) Central Ground Water Authority
- d) National Water Development Agency

5] With reference to Indian laws about wildlife protection, consider the following statements: [2022]

1. Wild animals are the sole property of the government.
2. When a wild animal is declared protected, such animal is entitled for equal protection whether it is found in protected areas or outside.
3. Apprehension of a protected wild animal becoming a danger to human life is sufficient ground for its capture or killing.

Which of the statements given above is/are correct?

- a) 1 and 2
- b) 2 only
- c) 1 and 3
- d) 3 only

6] With reference to India's Desert National Park, which of the following statements are correct? [2020]

1. It is spread over two districts.
2. There is no human habitation inside the Park.
3. It is one of the natural habitats of the Great Indian Bustard.

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

7] Among the following Tiger Reserves, which one has the largest area under 'Critical Tiger Habitat'? [2020]

- a) Corbett
- b) Ranthambore
- c) Nagarjunasagar-Srisailem
- d) Sundarbans

8] If a particular plant species is placed under Schedule VI of The Wildlife Protection Act, 1972, what is the implication? [2020]

- a) A licence is required to cultivate that plant.
- b) Such a plant cannot be cultivated under any circumstances.
- c) It is a Genetically Modified crop plant.
- d) Such a plant is invasive and harmful to the ecosystem.

9] Which of the following Protected Areas are located in Cauvery basin? [2020]

1. Nagarhole National Park
2. Papikonda National Park
3. Sathyamangalam Tiger Reserve
4. Wayanad Wildlife Sanctuary

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 3 and 4 only
- c) 1, 3 and 4 only
- d) 1, 2, 3 and 4

10] Which one of the following protected areas is well-known for the conservation of a sub-species of the Indian swamp deer (Barasingha) that thrives well on hard ground and is exclusively graminivorous? [2020]

- a) Kanha National Park
- b) Manas National Park
- c) Mudumalai Wildlife Sanctuary
- d) Tal Chhapar Wildlife Sanctuary

11] Which one of the following National Parks lies completely in the temperate alpine zone? [2019]

- a) Manas National Park
- b) Namdapha National Park
- c) Neora Valley National Park
- d) Valley of Flowers National Park

12] Which of the following are in Agasthyamala Biosphere Reserve? [2019]

- a) Neyyar, Idl Peppara and Shendurney Wildlife Sanctuaries; and Kalakad Mundanthurai Tiger Reserve

- b) Mudumalai, Sathyamangalam and Wayanad Wildlife Sanctuaries; and Silent Valley National Park
- c) Kaundinya, Gundla Brahmeswaram and Papikonda Wildlife Sanctuaries; and Mukurthi National Park
- d) Kawal and Sri Venkateswara Wildlife Sanctuaries; and Nagarjunasagar-Srisailem Tiger Reserve

13] Consider following statements: [2019]

1. Under Ramsar Convention, it is mandatory on the part of the Government of India to protect and conserve all the wetlands in the territory of India.
2. The Wetlands (Conservation and Management) Rules, 2010 were framed by the Government of India based on the recommendations of Ramsar Convention.
3. The Wetlands (Conservation and Management) Rules, 2010 also encompass the drainage area or catchment regions of the wetlands as determined by the authority.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

14] Consider the following statements: [2019]

1. As per law, the Compensatory Afforestation Fund Management and Planning Authority exists at both National and State levels.
2. People's participation is mandatory in the compensatory afforestation programmes carried out under the Compensatory Afforestation Fund Act, 2016.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

15] In which of the following States is Pakhui Wildlife Sanctuary located? [2018]

- a) Arunachal Pradesh
- b) Manipur
- c) Meghalaya
- d) Nagaland

16] Consider of the following statements: [2018]

1. The Definition of 'Critical Wildlife Habitat' is incorporated in the Forest Rights, 2006.
2. For the first time in India, Baigas have been given Habitat Rights.

3. Union Ministry of Environment, Forest and Climate Change officially decides and declares Habitat Rights for Primitives and Vulnerable Tribal Groups is any part of India.

Which of the statements given below is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

17] According to the Wildlife (Protection) Act, 1972, which of the following animals cannot be hunted by any person except under some provisions provided by law? [2017]

1. Gharial
2. Indian wild ass
3. Wild buffalo

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

18] The term 'M-STripES' is sometimes seen in the news in the context of: [2017]

- a) Captive breeding of Wild Fauna
- b) Maintenance of Tiger Reserves
- c) Indigenous Satellite Navigation System
- d) Security of National Highways

19] In India, if a species of tortoise is declared protected under Schedule I of the Wildlife (Protection) Act, 1972, what does it imply? [2017]

- a) It enjoys the same level of protection as the tiger.
- b) It no longer exists in the wild, a few individuals are under captive protection; and now it is impossible to prevent its extinction.
- c) It is endemic to a particular region of India.
- d) Both (b) and (c) stated above are correct in this context.

20] Recently there was a proposal to translocate some of the lions from their natural habitat in Gujarat to which one of the following sites? [2017]

- a) Corbett National Park
- b) Kuno Palpur Wildlife Sanctuary
- c) Mudumalai Wildlife Sanctuary
- d) Sariska National Park

21] From the ecological point of view, which one of the following assumes importance in being a good link between the Eastern Ghats and the Western Ghats? [2017]

- a) Sathyamangalam Tiger Reserve
- b) Nallamala Forest
- c) Nagarhole National Park
- d) Seshachalam Biosphere Reserve

22] Which of the following are the key features of 'National Ganga River Basin Authority (NGRBA)? [2016]

1. River basin is the unit of planning and management.
2. It spearheads the river conservation efforts at the national level.
3. One of the Chief Ministers of the States through which the Ganga flows becomes the Chairman of NGRBA on rotation basis.

Select the correct answer using the code given Below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

23] Which of the following best describes/ describe the aim of 'Green India Mission' of the Government of India? [2016]

1. Incorporating environmental benefits and costs into the Union and State Budgets thereby implementing the 'green accounting'
2. Launching the second green revolution to enhance agricultural output so as to ensure food security to one and all in the future
3. Restoring and enhancing forest cover and responding to climate change by a combination of adaptation and mitigation measures

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

24] Which one of the following National Parks has a climate that varies from tropical to subtropical, temperate and arctic? [2015]

- a) Khangchendzonga National park
- b) Nandadevi National Park

- c) Neora Valley National Park
- d) Namdapha National park

25] Which of the following National Parks is unique in being a swamp with floating vegetation that supports a rich biodiversity? [2015]

- a) Bhitarkanika National Park
- b) Keibul Lamjao National Park
- c) Keoladeo Ghana National park
- d) Sultanpur National park

26] With reference to 'Eco-Sensitive Zones', which of the following statements is/are correct? [2014]

1. Eco-Sensitive Zones are the areas that are declared under the Wildlife (Protection) Act, 1972.
2. The purpose of the declaration of EcoSensitive Zones is to prohibit all kinds of human activities in those zones except agriculture.

Select the correct answer using the code given below.

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

27] Consider the following statements: [2014]

1. Animal Welfare Board of India is established under the Environment (Protection) Act, 1986.
2. National Tiger Conservation Authority is a statutory body.
3. National Ganga River Basin Authority is chaired by the Prime Minister.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 2 only
- d) 1, 2 and 3

28] Consider the following pairs: [2014]

1. Dampa Tiger Reserve: Mizoram
2. Gumti Wildlife Sanctuary: Sikkim
3. Saramati Peak: Nagaland

Which of the above pairs is/are correctly matched?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

29] Consider the following pairs: [2014]

Wetlands: Confluence of rivers:

1. Harike Wetlands: Confluence of Beas and Satluj/ Suttlej

2. Keoladeo Ghana National Park: Confluence of Banas and Chambal

3. Kolleru Lake: Confluence of Musi and Krishna

Which of the above pairs is/are correctly matched?

- a) 1 only b) 2 and 3 only
c) 1 and 3 only d) 1, 2 and 3

30] What are the benefits of implementing the 'Integrated Watershed Development Programme' [2014]

1. Prevention of soil runoff
2. Linking the country's perennial rivers with seasonal rivers
3. Rainwater harvesting and recharge of groundwater table
4. Regeneration of natural vegetation

Select the correct answer using the code given below:

- a) 1 and 2 only b) 2, 3 and 4 only
c) 1, 3 and 4 only d) 1, 2, 3 and 4

31] Consider the following pairs: [2013]

National park: River flowing through the park

1. Corbett national park: Ganga
2. Kaziranga national park: Manas
3. Silent valley national park: Kaveri

Which of the above pairs is/are correctly matched?

- a) 1 and 2 b) 3 only
c) 1 and 3 d) None

32] Consider the following pairs : [2013]

1. Nokrek biosphere reserve : Garo Hills
2. Logtak (Loktak) lake : Bakail Range
3. Namdapha national park : Dafla Hills

Which of the above pairs is/are correctly matched?

- a) 1 only b) 2 and 3 only
c) 1, 2 and 3 d) None

33] With reference to the wetlands of India, consider the following statements: [2012]

1. The country's total geographical area under the category of wetlands is recorded more in Gujarat as compared to other States.
2. In India, the total geographical area of coastal wetlands is larger than that of inland wetlands.

Which of the statements given above is/are correct?

- a) Only 1 is correct b) Only 2 is correct
c) Both 1 & 2 d) Neither 1 nor 2

34] Consider the following protected areas:: [2012]

1. Bandipur
2. Bhitarkanika
3. Manas
4. Sunderbans

Which of the above are declared Tiger Reserves?

- a) 1 & 2 Only b) 1, 3 and 4 Only
c) 2, 3 and 4 only d) 1, 2, 3 & 4

35] How does National Biodiversity Authority (NBA) help in protecting the Indian agriculture? [2012]

1. NBA checks the biopiracy and protects the indigenous and traditional genetic resources.
2. NBA directly monitors and supervises the scientific research on genetic modification of crop plants.
3. Application for Intellectual Property Rights related to resources genetic/biological cannot be made without approval of NBA.

Which of the statements given above is/are correct?

- a) 1 Only b) 2 & 3 Only
c) 1 & 3 Only d) 1, 2 & 3

36] In which one among the following categories of protected areas in India are local people are not allowed to collect and use the biomass? [2012]

- a) Biosphere Reserves
b) National Parks
c) Wetlands declared under Ramsar Convention
d) Wildlife Sanctuaries

37] With reference to India, consider the following central acts: [2011]

1. Import and export (control) act, 1947.
2. Mining, and mineral development (regulation) act, 1957.
3. Customs act, 1962.
4. Indian forest act, 1927.

Which of the above acts have relevance to/bearing on the biodiversity conservation in the country?

- a) 1 and 3 only

- b) 2,3 and 4 only
- c) 1,2,3 and 4
- d) None of the above acts

38] Consider the following pairs:

Protected Area: Well Known for: [2010]

- 1. Bhitarkanika, Orissa- Salt Water Crocodile
- 2. Desert National Park, Rajasthan - Great Indian Bustard
- 3. Eravikulam, Kerala- Hootak Gibbon

Which of the pairs given above is/are correctly matched?

- a) 1 only
- b) 1 and 2 only
- c) 2 only
- d) 1, 2 and 3

39] India is a party to the Ramsar Convention and as declared many areas as Ramsar Sites. Which of the following statements best describes as to how we should maintain these sites in the context of this Convention? [2010]

- a) Keep all the sites completely inaccessible to man so that they will not be exploited.
- b) Conserve all the sites through ecosystem approach and permit tourism and recreation only.
- c) Conserve all the sites through ecosystem approach for a period without any exploitation, with specific criteria and specific period for each site, and then allow sustainable use of them by future generations.
- d) Conserve all the sites through ecosystem approach and allow their simultaneous and sustainable use.

40] Consider the following statements: [2010]

- 1. The boundaries of a National Park are defined by legislation.
- 2. A Biosphere Reserve is declared to conserve a few specific species of flora and fauna.
- 3. In a Wildlife Sanctuary, limited biotic interference is permitted.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

41] What are the possible limitations of India in mitigating the global warming at present and in immediate future? [2010]

- 1. Appropriate alternate technologies are not sufficiently available.
- 2. India cannot invest huge funds in research and development.
- 3. Many developed countries have already set up their polluting industries in India.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 1, 2 and 3

42] Which one of the following Union Ministries implements the Cartagena Protocol on Biosafety? [2009]

- a) Ministry of Science and Technology
- b) Ministry of Health and Family Welfare
- c) Ministry of Environment and Forests
- d) Ministry of Chemicals and Fertilizers

43] In India, which one of the following States has the largest inland saline wetland? [2009]

- a) Gujarat
- b) Haryana
- c) Madhya Pradesh
- d) Rajasthan

44] Where is the headquarters of Animal Welfare Board of India located? [2009]

- a) Ahmedabad
- b) Chennai
- c) Hyderabad
- d) Kolkata

45] Which one among the following has the maximum number of National Parks? [2008]

- a) Andaman and Nicobar Islands
- b) Arunachal Pradesh
- c) Assam
- d) Meghalaya

46] Out of all the Biosphere Reserves in India. Four have recognized on the World Network by UNESCO. Which one of the following is not one of them? [2008]

- a) Gulf of Mannar
- b) Knachejunga
- c) Nanda Devi
- d) Sunderbans

47] Which one of the following is located in the Bastar region? [2007]

- a) Bandhavgarh National Park

- b) Dundeli Sanctuary
- c) Rajai National Park
- d) Indravati National Park

48] Which one of the following is also known as Top Slip? [2007]

- a) Simlipal National Park
- b) Periyar Wildlife Sanctuary
- c) Manjira Wildlife Sanctuary
- d) Indira Gandhi Wildlife Sanctuary and National Park

49] Match List I with List II and select the correct answer using the code given below the lists: [2006]

List . I (National Park/ Wildlife Sanctuary)	List . II (Nearby Town)
A. Chandra Prabha	1. Jaipur
B. Karera	2. Jhansi
C. Jaisamand	3. Agra
D. Nahargarh	4. Varanasi
	5. Udaipur

- A B C D
- a) 4 1 5 2
 - b) 5 2 3 1
 - c) 4 2 5 1
 - d) 5 1 3 2

50] Where is Bundala Biosphere Reserve which has been recently added to the UNESCO's Man and Biosphere (MAB) network, located? [2006]

- a) Russia
- b) India
- c) Sri Lanka
- d) Bangladesh

51] Which one of the following is not a Biosphere Reserve? [2005]

- a) Agasthyamalai
- b) Nallamalai
- c) Nilgiri
- d) Panchmarhi

52] Match List-I (National Park/Wildlife Sanctuary) with List-II (State) and select the correct answer using the code given below the list: [2005]

List I	List II
A) Bondla Wildlife Sanctuary	1) Orissa
B) Kangerghat National Park	2) Assam
C) Orang Sanctuary	3) Chhattisgarh

- D) Ushakothi Wildlife Sanctuary
- 4) Goa
- 5) Tripura

- A B C D
- a) 2 1 5 3
 - b) 4 3 2 1
 - c) 2 3 5 1
 - d) 4 1 2 3

53] Match List-I (National Park/Sanctuary) with List-II (state) and select the correct answer using the codes given below the Lists: [2004]

List-I (National Park Sanctuary)	List-II (State)
A. Kanger Ghati National Park	1. Chhattisgarh
B. Nagerhole National Park	2. Haryana
C. Kugti Wildlife Sanctuary	3. Himachal Pradesh
D. Sultanpur Bird Sanctuary	4. Karnataka

- A B C D
- a) 3 2 1 4
 - b) 1 4 3 2
 - c) 3 4 1 2
 - d) 1 2 3 4

54] Match List-I (Biosphere Reserves) with List-II (States) and: [2004]

select the correct answer using the codes given below the Lists:

List-I (Biosphere Reserves)	List-II (States)
A. Similipal	1. Sikkim
B. Dehong Deband	2. Uttaranchal
C. Nokrek	3. Arunachal Pradesh
D. Kanchenjunga	4. Orissa
	5. Meghalaya

- A B C D
- a) 1 3 5 4
 - b) 4 5 2 1
 - c) 1 5 2 4
 - d) 4 3 5 1

55] Consider the following statements regarding environment issues of India: [2001]

- I. Gulf of Manner is one of the biosphere reserves.
- II. The Ganga Action Plan, phase II has been merged with the National River Conservation Plan.

a) The Water (Cess) Act, 1977
b) The Forest (Conservation) Act, 1980

a) Great Nicobar b) Sunderbans
c) Nanda Devi d) Gulf of Kachchh

61
d

F) Global Initiatives for Environment Protection

Global Initiatives for Climate, Environment, Biodiversity Protection etc.

1] Consider the following statements: [2025]

Statement I: Article 6 of the Paris Agreement on climate change is frequently discussed in global discussions on sustainable development and climate change.

Statement II: Article 6 of the Paris Agreement on climate change sets out the principles of carbon markets.

Statement III: Article 6 of the Paris Agreement on climate change intends to promote inter-country non-market strategies to reach their climate targets. Which one of the following is correct in respect of the above statements?

- a) Both Statement II and Statement III are correct and both of them explain Statement I
- b) Both Statement II and Statement III are correct but only one of them explains Statement I
- c) Only one of the Statements II and III is correct and that explains Statement I
- d) Neither Statement II nor Statement III is correct

2] Consider the following statements: [2025]

Statement I: At the 28th United Nations Climate Change Conference (COP28), India refrained from signing the “Declaration on Climate and Health”.

Statement II: The COP28 Declaration on Climate and Health is a binding declaration; and if signed, it becomes mandatory to decarbonize health sector.

Statement III: If India’s health sector is decarbonized, the resilience of its health-care system may be compromised.

Which one of the following is correct in respect of the above statements?

- a) Both Statement II and Statement III are correct and both of them explain Statement I
- b) Both Statement II and Statement III are correct but only one of them explains Statement I
- c) Only one of the Statements II and III is correct and that explains Statement I
- d) Neither Statement II nor Statement III is correct

3] Which one of the following launched the 'Nature Solutions Finance Hub for Asia and the Pacific'? [2025]

- a) The Asian Development Bank (ADB)
- b) The Asian Infrastructure Investment Bank (AIIB)
- c) The New Development Bank (NDB)
- d) The International Bank for Reconstruction and Development (IBRD)

4] Which organization has enacted the Nature Restoration Law (NRL) to tackle climate change and biodiversity loss? [2025]

- a) The European Union
- b) The World Bank
- c) The Organization for Economic Cooperation and Development
- d) The Food and Agriculture Organization

5] Consider the following statements: [2024]

Statement I: The European Parliament approved the Net-Zero Industry Act recently

Statement-II: The European Union intends to achieve carbon neutrality by 2040 and therefore aims to develop all of its own clean technology by that time. Which one of the following is correct in respect of the above statements?

- (a) Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

6] ‘Invasive Species Specialist Group’ (that develops Global Invasive Species Database) belongs to which one of the following organizations? [2023]

- a) The International Union for Conservation of Nature
- b) The United Nations Environment Programme
- c) The United Nations World Commission for Environment and Development
- d) The World-Wide Fund for Nature

7] Consider the following statements: [2023]

Statement-I: Carbon markets are likely to be one of the most widespread tools in the fight against climate change.

Statement-II: Carbon market transfer resources from private sector to the State.

Which of the following is correct in respect of the above statements?

- a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I.
- b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I.
- c) Statement-I is correct but Statement-II is incorrect.
- d) Statement-I is incorrect but Statement-II is correct.

8] "Climate Action Tracker" which monitors the emission reduction pledges of different countries is a: [2022]

- a) Database created by coalition of research organisations
- b) Wing of "International Panel of Climate Change"
- c) Committee under "United Nations Framework Convention on Climate Change"
- d) Agency promoted and financed by United Nations Environment Programme and World Bank

9] Consider the following statements: [2022]

- 1. "The Climate Group" is an international non-profit organization that drives climate action by building large networks and runs them.
- 2. The International Energy Agency in partnership with the Climate Group launched a global initiative "EP100".
- 3. EP100 brings" together leading companies committed to driving innovation in energy efficiency and increasing competitiveness while delivering on emission reduction goals.
- 4. Some Indian companies are members of EP100.
- 5. The International Energy Agency is the Secretariat to the "Under2 Coalition".

Which of the statements given above are Correct?

- a) 1,2,4 and5
- b) 1, 3 and 4 only

- c) 2, 3 and 5 only
- d) 1, 2, 3, 4 and 5

10] The "Miyawaki method" is well known for the: [2022]

- a) Promotion of commercial farming in arid and semi-arid areas
- b) Development of gardens using genetically modified flora
- c) Creation of mini forests in urban areas
- d) Harvesting wind energy on coastal areas and on sea surfaces

11] The 'Common Carbon Metric', supported by UNEP, has been developed for: [2021]

- a) Assessing the carbon footprint of building operations around the world
- b) Enabling commercial farming entities around the world to enter carbon emission trading
- c) Enabling governments to assess the overall carbon footprint caused by their countries
- d) Assessing the overall carbon footprint caused by the use of fossil fuels by the world in a unit time

12] With reference to the New York declaration on Forests, which of the following statements are correct? [2021]

- 1) It was first endorsed at the United Nations Climate Summit in 2014.
 - 2) It endorses a global timeline to end the loss of forests.
 - 3) It is a legally binding international declaration.
 - 4) It is endorsed by governments, big companies and indigenous communities.
 - 5) India was one of the signatories at its inception.
- Select the correct answer using the code given below.

- a) 1,2 and 4
- b) 1, 3 and 5
- c) 3 and 4
- d) 2 and 5

13] 'Momentum for Change: Climate Neutral Now' is an initiative launched by: [2018]

- a) The Intergovernmental Panel on Climate Change
- b) The UNEP Secretariat
- c) The UNFCCC Secretariat
- d) The World Meteorological Organization

14] The Partnership for Action Green Economy (PAGE), a UN mechanism to assist countries transition towards greener and more inclusive economies, emerged at: [2018]

- a) The Earth Summit on Sustainable Development 2002, Johannesburg
- b) The United Nations Conference on Sustainable Development 2012, Rio De Janeiro
- c) The United Nations Framework Convention on Climate Change 2015, Paris
- d) The World Sustainable Development Summit 2016, New Delhi

15] With reference to 'Global Climate Change Alliance', which of the following statements is/are correct? [2017]

- 1. It is an initiative of the European Union.
- 2. It provides technical and financial support to targeted developing countries to integrate climate change into their development policies and budgets.
- 3. It is coordinated by World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD).

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 3 only
- c) 2 and 3 only
- d) 1, 2 and 3

16] Consider the following statements: [2017]

- 1. Climate and Clean Air Coalition (CCAC) to Reduce Short Lived Climate Pollutants is a unique initiative of G20 group of countries.
- 2. The CCAC focuses on methane, black carbon and hydrofluorocarbons.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

17] Consider the following statements in respect of Trade Related Analysis of Fauna and Flora in Commerce (TRAFFIC): [2017]

- 1. TRAFFIC is a bureau under United Nations Environment Programme (UNEP).

2. The mission of TRAFFIC is to ensure that trade in wild plants and animals is not a threat to the conservation of nature.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

18] With reference to an initiative called 'The Economics of Ecosystems and Biodiversity (TEEB)', which of the following statements is/are correct? [2016]

- 1. It is an initiative hosted by UNEP, IMF and World Economic Forum.
- 2. It is a global initiative that focuses on drawing attention to the economic benefits of biodiversity.
- 3. It presents an approach that can help decision-makers recognize, demonstrate and capture the value of ecosystems and biodiversity.

Select the correct answer using the code given below.

- a) 1 and 2 only
- b) 3 only
- c) 2 and 3 only
- d) 1, 2 and 3

19] Which of the following statements is/are correct? [2016]

Proper design and effective implementation of UN-REDD+ Programme can significantly contribute to

- 1. Protection of biodiversity
- 2. Resilience of forest ecosystems
- 3. Poverty reduction

Select the correct answer using the code given below.

- a) 1 and 2 only
- b) 3 only
- c) 2 and 3 only
- d) 1, 2 and 3

20] Consider the following pairs: [2016]

Terms sometimes seen in the news

Their origin

- 1. Annex-I Countries : Cartagena Protocol
 - 2. Certified Emissions Reductions: Nagoya Protocol
 - 3. Clean Development Mechanism: Kyoto Protocol
- Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

21] What is 'Greenhouse Gas Protocol'? [2016]

- a) It is an international accounting tool for government and business leaders to understand, quantify and manage greenhouse gas emissions
- b) It is an initiative of the United Nations to offer financial incentives to developing countries to reduce greenhouse gas emissions and to adopt eco-friendly technologies
- c) It is an inter-governmental agreement ratified by all the member countries of the United Nations to reduce greenhouse gas emissions to specified levels by the year 2022
- d) It is one of the multilateral REDD+ initiatives hosted by the World Bank

22] With reference to 'Agenda 21', sometimes seen in the news, consider the following statements: [2016]

- 1. It is a global action plan for sustainable development
- 2. It originated in the World Summit on Sustainable Development held in Johannesburg in 2002.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

23] With reference to the Agreement at the UNFCCC Meeting in Paris in 2015, which of the following statements is/are correct? [2016]

- 1. The Agreement was signed by all the member countries of the UN and it will go into effect in 2017.
- 2. The Agreement aims to limit the greenhouse gas emissions so that the rise in average global temperature by the end of this century does not exceed 2°C or even 1.5°C above pre-industrial levels.
- 3. Developed countries acknowledged their historical responsibility in global warming and committed to donate \$ 1000 billion a year from 2020 to help developing countries to cope with climate change.

Select the correct answer using the code given below.

- a) 1 and 3 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

24] Consider the following statements: [2016]

1. The Sustainable Development Goals were first proposed in 1972 by a global think tank called the 'Club of Rome'.

2. The Sustainable Development Goals have to be achieved by 2030.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

25] The term 'Intended Nationally Determined Contributions' is sometimes seen in the news in the context of: [2016]

a) pledges made by the European countries to rehabilitate refugees from the war-affected Middle East

b) plan of action outlined by the countries of the world to combat climate change

c) capital contributed by the member countries in the establishment of Asian Infrastructure Investment Bank

d) plan of action outlined by the countries of the world regarding Sustainable Development Goals

26] 'BioCarbon Fund Initiative for Sustainable Forest Landscapes' is managed by the: [2015]

a) Asian Development Bank

b) International Monetary Fund

c) United Nations Environment Programme

d) World Bank

27] With reference to 'Forest Carbon Partnership Facility', which of the following statements is/are correct? [2015]

1. It is global partnership of governments, businesses, civil society and indigenous peoples

2. It provides financial aid to universities, individual scientists and institutions involved in scientific forestry research to develop eco-friendly and climate adaptation technologies for sustainable forest management

3. It assists the countries in their 'REDD+ (Reducing Emission from Deforestation and Forest Degradation+)' efforts by providing them with financial and technical assistance.

Select the correct answer using the code given below

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

28] Which one of the following is associated with the issue of control and phasing out of the use of ozone-depleting substances? [2015]

- a) Bretton Woods Conference
- b) Montreal Protocol
- c) Kyoto Protocol
- d) Nagoya Protocol

29] What is Rio+20 Conference, often mentioned in the news? [2015]

- a) It is the United Nations Conference on Sustainable Development
- b) It is a Ministerial Meeting of the World Trade Organization
- c) It is a Conference of the Inter-governmental Panel on Climate Change
- d) It is a Conference of the Member Countries of the Convention on Biological Diversity

30] Which of the following statements regarding 'Green Climate Fund' is/are correct? [2015]

1. It is intended to assist the developing countries in adaptation and mitigation practices to counter climate change.
2. It is founded under the aegis of UNEP, OECD, Asian Development Bank and World Bank

Select the correct answer using the code given below.

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

31] With reference to 'Global Environment Facility', which of the following statements is/are correct? [2014]

- a) It serves as financial mechanism for 'Convention on Biological Diversity' and 'United Nations Framework Convention on Climate Change'.
- b) It undertakes scientific research on environmental issues at global level.
- c) It is an agency under OECD to facilitate the transfer of technology and funds to underdeveloped countries with specific aim to protect their environment.

- d) Both (a) and (b)

32] Regarding 'carbon credits' , which one of the following statements is not correct ? [2011]

- a) The carbon credit system was ratified in conjunction with the Kyoto protocol.
- b) Carbon credits are awarded to countries or groups that have reduced greenhouse gases below their emission quota.
- c) The goal of the carbon credit system is to limit the increase of carbon dioxide emission.
- d) Carbon credits are traded at a price fixed from time to time by the United Nations environment programs.

33] As a result of their annual survey, the National Geographic Society and an international polling firm Globe Scan gave India top rank in Greendex 2009 score. What is this score? [2010]

- a) It is a measure of efforts made by different countries in adopting technologies for reducing carbon footprint?
- b) It is a measure of environmentally sustainable consumer behaviour in different countries.
- c) It is an assessment of programmes / schemes undertaken by different countries for improving the conservation of natural resources.
- d) It is an index showing the volume of carbon credits sold by different countries.

34] The United Nations Framework Conventions on Climate Change (UNFCCC) is an international treaty drawn at: [2010]

- a) United Nations Conference on the Human Environment, Stockholm, 1972.
- b) UN conference on Environment and Development, Rio de Janeiro, 1992.
- c) World Summit on Sustainable Development, Johannesburg, 2002.
- d) UN Climate Change Conference Copenhagen, 2009.

35] Sustainable development is described as the development that meets the needs of the present without compromising the ability of future

generations to meet their own needs. In this perspective, inherently the concept of sustainable development is intertwined with which of the following concepts? [2010]

- a) Social justice and empowerments
- b) Inclusive Growth
- c) Globalization
- d) Carrying capacity

36] The concept of carbon credit originated from which one of the following? [2009]

- a) Earth Summit, Rio de Janeiro
- b) Kyoto Protocol
- c) Montreal Protocol
- d) G- 8 Summit, Heiligendamm

37] In the context of CO₂ emission and Global Warming, what is the name of a market driven device under the UNFCCC that allows developing countries to get funds/incentives from the developed countries to adopt better technologies that reduce greenhouse gas emissions? [2009]

- a) Carbon Footprint
- b) Carbon Credit Rating
- c) Clean Development Mechanism
- d) Emission Reduction Norm

38] Where was the World Summit on Sustainable Development (Rio+10) held? [2008]

- a) Davos
- b) Nova Scotia
- c) Johannesburg
- d) Shanghai

39] Consider the following statements: [2008]

1. Clean Development Mechanism (CDM) in respect of carbon credits in one of the Kyoto Protocol Mechanisms.
2. Under the CDM, the projects handled pertain only to the Annex-I countries.

Which of the statements given above is/are correct?

- a) 1 Only
- b) 2 Only
- c) Both 1 and 2
- d) Neither 1 nor 2

40] Which one of the following countries is the first country in the world to propose a carbon tax for its people to the address global warming? [2006]

- a) Australia
- b) Germany
- c) Japan
- d) New Zealand

41] Which of the statements is correct? [2005]

- a) The first meeting of the Parties (MOP 1) to the Cartagena Protocol on Biosafety was held in Philippines in the year 2004.
- b) India is not a signatory to the Biosafety Protocol/Convention on Biological Diversity.
- c) The Biosafety Protocol deals with genetically modified organisms.
- d) The United States of America is member of the Biosafety Protocol/ Convention on Biological Diversity.

42] Consider the following statements: [2005]

1. Kyoto Protocol came into force in the year 2005.
2. Kyoto Protocol deals primarily with the depletion of the Ozone layer.
3. Methane as a greenhouse gas is more harmful than carbon dioxide.

Which of the statements is/are correct?

- a) 1 and 2
- b) 1 and 3
- c) 1 only
- d) 3 only

43] A World Summit with representation from all the countries will be held in Johannesburg in September 2002 on the subject of: [2002]

- a) AIDS control
- b) Global terrorism
- c) Human rights
- d) Sustainable development

44] According to Meadows (1972), if the present trends in world population, industrialisation, pollution, food production and resource depletion continue unchanged, the 'Limits to Growth' on our planet will be reached in the next: [1998]

- a) 50 years
- b) 100 years
- c) 150 years
- d) 200 years

45] World Environment Conference to discuss global warming was held in 1997 in: [1998]

- a) Stockholm
- b) Rio de Janeiro
- c) Paris
- d) Kyoto

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
a	c	a	a	c	a	b	a	b	c	a	a	c	b	a	b	b	c	d	c

21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
a	a	b	b	b	d	c	b	a	a	a	d	b	b	d	b	c	c	a	d

41	42	43	44	45
c	b	d	b	d

G) Agriculture and Environment

1] Which of the following are nitrogen-fixing plants? [2022]

1. Alfalfa
2. Amaranth
3. Chickpea
4. Clover
5. Purslane (Kulfa)
6. Spinach

Select the correct answer using the code given below:

- a) 1, 3 and 4 only b) 1, 3, 5 and 6 only
c) 2, 4, 5 and 6 only d) 1, 2, 4, 5 and 6

2] "System of Rice Intensification" of cultivation, in which alternate wetting and drying of rice fields is practised, results in: [2022]

1. Reduced seed requirement
2. Reduced methane production
3. Reduced electricity consumption

Select the correct answer using the code given below:

- a) 1 and 2 only b) 2 and 3 only
c) 1 and 3 only d) 1, 2 and 3

3] In the context of India's preparation for Climate-Smart Agriculture, consider the following statements: [2021]

- 1) The 'Climate-Smart Village' approach in India is a part of a project led by the Climate Change, Agriculture and Food Security (CCAFS), an international research programme.

- 2) The project of CCAFS is carried out under Consultative Group on International Agricultural Research (CGIAR) headquartered in France.

- 3) The International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) in India is one of the CGIAR's research centres.

Which of the statements given above are correct?

- a) 1 and 2 only b) 2 and 3 only
c) 1 and 3 only d) 1, 2 and 3

4] What is/are the advantage/advantages of zero tillage in agriculture? [2020]

1. Sowing of wheat without burning the residue of previous crop.
2. Without the need for nursery of rice saplings, direct planting of paddy seeds in the wet soil is possible.
3. Carbon sequestration in the soil is possible

Select the correct answer using the code given below:

- a) 1 and 2 only b) 2 and 3 only
c) 3 only d) 1, 2 and 3

5] In the context of India, which of the following is/are considered to be practice(s) of eco-friendly agriculture? [2020]

1. Crop diversification
2. Legume intensification
3. Tensiometer use
4. Vertical farming

Select the correct answer using the code given below:

- a) 1, 2 and 3 only b) 3 only
c) 4 only d) 1, 2, 3 and 4

6] What is the use of biochar in farming? [2020]

1. Biochar can be used as a part of the growing medium in vertical farming.
2. When biochar is a part of the growing medium, it promotes the growth of nitrogen-fixing microorganisms.
3. When biochar is a part of the growing medium, it enables the growing medium to retain water for longer time.

Which of the statements given above is/are correct?

- a) 1 and 2 only b) 2 only
c) 1 and 3 only d) 1, 2 and 3

7] With reference to the 'Global Alliance for Climate Smart Agriculture (GACSA)', which of the following statements is/are correct? [2018]

1. GACSA is an outcome of the climate summit held in Paris in 2015.
2. Membership of GACSA does not create any binding obligations.
3. India was instrumental in the creation of GACSA.

Select the correct answer using the code given below:

- a) 1 and 3 only b) 2 only
c) 2 and 3 only d) 1, 2 and 3

8] With references to organic farming in India, consider the following statements: [2018]

1. 'The National Programme for Organic Production'(NPOP) is operated under the guidelines and directions of the Union Ministry Of the Rural Development.
2. 'The Agricultural and Processed and Products Export Development Authority'(APEDA) functions as the Secretariat for the implementation of NPOP.
3. Sikkim has become India's first fully organic State.

Which of the following given above is/are correct?

- a) 1 and 2 only b) 2 and 3 only
c) 3 only d) 1, 2 and 3

9] Which of the following is/are the advantage /advantages of practising drip irrigation? [2016]

1. Reduction in weed
2. Reduction in soil salinity
3. Reduction in soil erosion

Select the correct answer using the code given below.

- a) 1 and 2 only
b) 3 only
c) 1 and 3 only
d) None of the above is an advantage of practising drip irrigation

10] Why does the Government of India promote the use of Neem-coated Urea' in agriculture? [2016]

- a) Release of Neem oil in the soil increases nitrogen fixation by the soil microorganisms
- b) Neem coating slows down the rate of dissolution of urea in the soil
- c) Nitrous oxide, which is a greenhouse gas, is not at all released into atmosphere by crop fields
- d) Six is a combination of a weedicide and a fertilizer for particular crops

11] What can be the impact of excessive/inappropriate use of nitrogenous fertilizers in agriculture? [2015]

1. Proliferation of nitrogen-fixing microorganisms in soil can occur.
2. Increase in the acidity of soil can take place
3. Leaching of nitrate to the ground-water can occur.

Select the correct answer using the code given below.

- a) 1 and 3 only b) 2 only
c) 2 and 3 only d) 1, 2 and 3

12] Consider the following statements: [2015]

1. The Accelerated Irrigation Benefits Programme was launched during 1996-97 to provide loan assistance to poor farmers.
2. The Command Area Development Programme was launched in 1974-75 for the development of water-use efficiency.

Which of the statements given above is/are correct?

- a) 1 only b) 2 only
c) Both 1 and 2 d) Neither 1 nor 2

13] Consider the following statements: [2014]

1. Maize can be used for the production of starch.
2. Oil extracted from maize can be a feedstock for biodiesel.
3. Alcoholic beverages can be produced by using maize.

Which of the statements given above is/are correct?

- a) 1 only b) 1 and 2 only
c) 2 and 3 only d) 1, 2 and 3

14] What are the significances of a practical approach to sugarcane production known as 'Sustainable Sugarcane Initiative'? [2014]

1. Seed cost is very low in this compared to the conventional method of cultivation.
2. Drip irrigation can be practiced very effectively in this.
3. There is no application of chemical/inorganic fertilizers at all in this.
4. The scope for intercropping is more in this compared to the conventional method of cultivation.

Select the correct answer using the code given below.

- a) 1 and 3 only b) 1, 2 and 4 only
c) 2, 3 and 4 only d) 1, 2, 3 and 4

15] Consider the following organisms: [2013]

1. Agaricus
2. Nostoc
3. Spirogyra

Which of the above is/are used as biofertilizer/ biofertilizers?

- a) 1 and 2 b) 2 only
c) 2 and 3 d) 3 only

16] Consider the following kinds of organisms: [2012]

1. Bacteria
2. Fungi
3. Flowering plants

Some species of which of the above kinds of organisms are employed as biopesticides?

- a) 1 Only b) 2 & 3 Only
c) 1 & 3 Only d) 1,2 & 3

17] With reference to micro-irrigation, which of the following statements is/are correct? [2011]

1. Fertilizer/nutrient loss can be reduced.
2. It is the only means of irrigation in dry land farming.
3. In some areas of farming, receding of ground water table can be checked.

Select the correct answer using the codes given below:

- a) 1 only b) 2 and 3 only
c) 1 and 3 only d) 1, 2 and 3

18] Salinization occurs when the irrigation water accumulated in the soil evapo-rates, leaving behind salts and minerals. What are the effects of Salinization on the irrigated land? [2011]

- a) It greatly increases the crop production.
- b) It makes some soils impermeable.
- c) It raises the water table.
- d) It fills the air spaces in the soil with water.

19] Which features of some species of blue- green algae helps promote them as bio- fertilizers? [2010]

- a) They convert atmospheric methane into ammonia which the crop plants can absorb readily.
- b) They induce the crop plants to produce the enzymes which help convert atmospheric nitrogen to nitrates.
- c) They have the mechanism to convert atmospheric nitrogen into a form that the crop plants can absorb readily.
- d) They induce the roots of the crop plants to absorb the soil nitrates in larger quantities.

20] Which one of the following agricultural practices is eco-friendly? [1999]

- a) Organic farming
- b) Shifting cultivation
- c) Cultivation of high-yielding varieties
- d) Growing plants in glass-houses

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
a	d	d	d	a	d	b	b	c	b	c	b	d	b	b	d	c	b	c	a

H) Renewable and Alternative Energy Sources

Renewable Energy Sources, Non-conventional Energy Sources, Solar Energy, Bio-fuels, New Inventions in the Field

1] Consider the following statements: [2025]

Statement I: Of the two major ethanol producers in the world, i.e., Brazil and the United States of America, the former produces more ethanol than the latter.

Statement II: Unlike in the United States of America where corn is the principal feedstock for ethanol production, sugarcane is the principal feedstock for ethanol production in Brazil.

Which of the statements given above are correct?

- a) Both Statement I and Statement II are correct and Statement II explains Statement I
- b) Both Statement I and Statement II are correct but Statement II does not explain Statement I
- c) Statement I is correct but Statement II is not correct
- d) Statement I is not correct but Statement II is correct

2] Consider the following types of vehicles: [2025]

I. Full battery electric vehicles

II. Hydrogen fuel cell vehicles

III. Fuel cell electric hybrid vehicles

How many of the above are considered as alternative (powertrain) vehicles?

- a) Only one
- b) Only two
- c) All the three
- d) None

3] Consider the following statements about 'PM Surya Ghar Muft Bijli Yojana': [2025]

I. It targets installation of one crore solar rooftop panels in the residential sector.

II. The Ministry of New and Renewable Energy aims to impart training on installation, operation, maintenance and repairs of solar rooftop systems at grassroot levels.

III. It aims to create more than three lakhs skilled manpower through fresh skilling, and up-skilling, under scheme component of capacity building.

Which of the statements given above are correct?

- a) I and II only
- b) I and III only
- c) II and III only
- d) I, II and III

4] Consider the following substances: [2025]

I. Ethanol

II. Nitroglycerine

III. Urea

Coal gasification technology can be used in the production of how many of them?

- a) Only one
- b) Only two
- c) All the three
- d) None

5] Consider the following: [2024]

1. Battery storage

2. Biomass generators

3. Fuel cells

4. Rooftop solar photovoltaic units

How many of the above are considered "Distributed Energy Resources"?

- a) Only one
- b) Only two
- c) Only three
- d) All four

6] Consider the following materials: [2024]

1. Agricultural residues

2. Corn grain

3. Wastewater treatment sludge

4. Wood mill waste

Which of the above can be used as feedstock for producing Sustainable Aviation Fuel?

- a) 1 and 2 only b) 3 and 4 only
- c) 1, 2, 3 and 4 d) 1, 3 and 4 only

7] One of the following is the exhaust pipe emission from Fuel Cell Electric Vehicles, powered by hydrogen? [2024]

- a) Hydrogen peroxide b) Hydronium
- c) Oxygen d) Water vapour

8] Recently, the term "pumped-storage hydropower" is actually and appropriately discussed in the context of which one of the following? [2024]

- (a) Irrigation of terraced crop fields
- (b) Lift irrigation of cereal crops
- (c) Long duration energy storage
- (d) Rainwater harvesting system

9] With reference to green hydrogen, consider the following statements: [2023]

- 1. It can be used directly as a fuel for internal combustion.
- 2. It can be blended with natural gas and used as fuel for heat or power generation.
- 3. It can be used in the hydrogen fuel cell to run vehicles.

How many of the above statements are correct?

- a) Only one b) Only two
- c) All three d) None

10] Consider the following heavy industries: [2023]

- 1. Fertilizer plants
- 2. Oil refineries
- 3. Steel plants

Green hydrogen is expected to play a significant role in decarbonizing how many of the above industries?

- a) Only one b) Only two
- c) All three d) None

11] Consider the following statements: [2022]

- 1. Gujarat has the largest solar park in India.

2. Kerala has a fully solar powered International Airport.

3. Goa has the largest floating solar photovoltaic project in India.

Which of the statements given below is/are correct?

- a) 1 and 2 b) 2 only
- c) 1 and 3 d) 3 only

12] According to India's National Policy on Biofuels, which of the following can be used as raw materials for the production of biofuels? [2020]

- 1. Cassava
- 2. Damaged wheat grains
- 3. Groundnut seeds
- 4. Horse gram
- 5. Rotten potatoes
- 6. Sugar beet

Select the correct answer using the code given below:

- a) 1, 2, 5 and 6 only b) 1, 3, 4 and 6 only
- c) 2, 3, 4 and 5 only d) 1, 2, 3, 4, 5 and 6

13] With reference to solar water pumps, consider the following statements: [2020]

- 1. Solar power can be used for running surface pumps and not for submersible pumps.
- 2. Solar power can be used for running centrifugal pumps and not the ones with piston.

Which of the statements given above is/are correct?

- a) 1 only b) 2 only
- c) Both 1 and 2 d) Neither 1 nor 2

14] It is possible to produce algae-based biofuels, but what is/are the likely limitation(s) of developing countries in promoting this industry? [2017]

- 1. Production of algae-based biofuels is possible in seas only and not on continents.
- 2. Setting up and engineering the algae-based biofuel production requires high level of expertise/technology until the construction is completed.
- 3. Economically viable production necessitates the setting up of large-scale facilities which may raise ecological and social concerns.

Select the correct answer using the code given below:

- a) 1 and 2 only b) 2 and 3 only
c) 3 only d) 1, 2 and 3

15] With reference to the Indian Renewable Energy Development Agency Limited (IREDA), which of the following statements is/are correct? [2015]

1. It is a Public Limited Government Company.
2. It is a Non ' Banking Financial Company.

Select the correct answer using the code given below.

- a) 1 only b) 2 only
c) Both 1 and 2 d) Neither 1 or 2

16] Biomass gasification is considered to be one of the sustainable solutions to the power crisis in India. In this context, which of the following statements is/are correct? [2012]

1. Coconut shells, groundnut shells and rice husk can be used in biomass gasification.
2. The combustible gases generated from biomass gasification consist of hydrogen and carbon dioxide only.
3. The combustible gases generated from biomass gasification can be used for direct heat generation but not in internal combustion engines.

Select the correct answer using the codes given below:

- a) 1 Only b) 2 & 3 Only
c) 1 & 3 Only d) 1,2 & 3

17] Microbial fuel cells are considered a source of sustainable energy. Why? [2011]

1. They use living organisms as catalysts to generate electricity from certain substrates.
2. They use a variety of inorganic materials as substrates.

3. They can be installed in waste water treatment plants to cleanse water and produce electricity.

Which of the following statements given above is/ are correct?

- a) 1 only b) 2 and 3 only
c) 1 and 3 only d) 1,2, and 3

18] Given below are the names of four energy crops. Which one of them can be cultivated for ethanol? [2010]

- a) Jatropha b) Maize
c) Pongamia d) Sunflower

19] Other than Jatropha curcas, why is Pongamia pinnata also considered a good option for the production of bio- diesel in India? [2010]

1. Pongamia Pinnata grows naturally in most of the arid regions of India.
2. The seeds of Pongamia pinnata are rich in lipid content of which nearly half is oleic acid.

Which of the statements given above is/are correct?

- a) 1 only b) 2 only
c) Both 1 and 2 d) Neither 1 nor 2

20] In the context of alternative sources of energy, ethanol as a viable bio-fuel can be obtained from: [2009]

- a) Potato b) Rice
c) Sugarcane d) Wheat

21] Which one of the following Union Ministries is implementing the Biodiesel Mission (as Noida Ministry)? [2008]

- a) Ministry of Agriculture
b) Ministry of Science and Technology
c) Ministry of New and Renewable Energy
d) Ministry of Rural Development

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
d	c	d	c	d	c	d	c	c	c	b	a	d	b	c	a	c	b	c	c

21
d